

TOPOLOGY

2026 Spring

Author: Tianyi Chen¹

¹chenty24@mails.tsinghua.edu.cn

Contents

0	Introduction	3
1	Fundamental group and covering space	4
1.1	Basis	4
1.2	Easy Result	5
1.3	Exercise	5
2	Homology	9
2.1	Basis	9
2.2	Easy Result	9
2.3	Exercise	10
3	Cohomology	20
3.1	Basis	20
3.2	Easy Result	21
3.3	Exercise	22
4	Homotopy theory	29
4.1	Basis	29
4.2	Easy Result	30
4.3	Exercise	31
5	Miscellaneous	37
5.1	Basis	37
5.2	Exercise	37

Chapter 0

Introduction

I was preparing Yau's contest and Qiuzhen qualifying exams in 2026 spring, and wrote this problem set about algebraic topology. There is no problem sets about algebraic topology right now, though many people will say Hatcher's book is enough. I still want to add more, so wrote this, leave something to help myself review and may help others who want to do a lot exercise on topology.

The problem set mainly contains exercise on Hatcher's book, Qiuzhen qualifying exams, and Yau's contest. I select some interesting and challenging exercise to write, and the ones which is easy to write answers(hhh). Since the time is limited, this problem set contains still a little exercise. And I will update from time to time, and add more exercise, on my github ' <https://tianyic089.github.io/> '.

The answers must have some typos or mistakes, or if you have questions on my answers, or you have some fantastic problem want to tell me, welcome to contact me in email. I will appreciate a lot.

Chapter 1

Fundamental group and covering space

1.1 Basis

Theorem 1.1.1 (Van Kampen theorem). *Suppose $X = A \cup B$, $A, B, A \cap B \subset X$ are all path-connected open subspaces. Choose a base point $x_0 \in A \cap B$. We have inclusions*

$$\begin{aligned} i_{A,B} : A, B &\hookrightarrow X \\ j_{A,B} : A \cap B &\hookrightarrow A, B \end{aligned}$$

then

$$i_{A*} * i_{B*} : \pi_1(A, x_0) * \pi_1(B, x_0) \rightarrow \pi_1(X, x_0)$$

is surjective with kernel

$$\ker h = [j_{A*}(\omega) * j_{B*}(\omega)^{-1} \mid \omega \in \pi_1(A \cap B, x_0)]$$

Remark 1.1.2. *The intuition is that $\pi_1(X, x_0)$ comes from $\pi_1(A, x_0) * \pi_1(B, x_0)$ by imposing the relation $\pi_1(A \cap B, x_0)$.*

Remark 1.1.3. *Van Kampen theorem for $X = \bigcup_{\alpha} A_{\alpha}$ can be found in [1] theorem 1.20.*

Method 1.1.4. *Computing π_1 for a CW complex can remove cells that $\dim > 2$.*

Corollary 1.1.5. *For every group G , there is a 2-dimensional CW complex X_G with $\pi_1(X_G) = G$.*

Proposition 1.1.6 (Homotopy lifting property). *Given a covering space $p : \tilde{X} \rightarrow X$, a homotopy $f_t : Y \rightarrow X$, and a map $\tilde{f}_0 : Y \rightarrow \tilde{X}$ lifting f_0 , then there exists a unique homotopy $\tilde{f}_t : Y \rightarrow \tilde{X}$ of $\tilde{f}_0$ that lifts f_t .*

Proposition 1.1.7. *The number of sheets of a covering space $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ with X and $\tilde{X}$ path-connected equals the index of $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ in $\pi_1(X, x_0)$.*

Proposition 1.1.8 (Lifting criterion). *Given a covering space $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ and a map $f : (Y, y_0) \rightarrow (X, x_0)$ with Y path-connected and locally path-connected. Then a lift $\tilde{f} : (Y, y_0) \rightarrow (\tilde{X}, \tilde{x}_0)$ exists iff $f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(\tilde{X}, \tilde{x}_0))$.*

Theorem 1.1.9 (Classification theorem). *Let X be path-connected, locally path-connected, and semilocally simply-connected. Then there is a bijection between the set of basepoint-preserving isomorphism classes of path-connected covering spaces $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ and the set of subgroups of $\pi_1(X, x_0)$.*

Proposition 1.1.10. *Let $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be a path-connected covering space of the path-connected, locally path-connected space X , and let H be the subgroup $p_*(\pi_1(\tilde{X}, \tilde{x}_0)) \subset \pi_1(X, x_0)$. Then:*

(a) *This covering space is normal iff H is a normal subgroup of $\pi_1(X, x_0)$.*

- (b) The group of deck transformation $G(\tilde{X})$ is isomorphic to the quotient $N(H)/H$ where $N(H)$ is the normalizer of H in $\pi_1(X, x_0)$.

In particular, $G(\tilde{X})$ is isomorphic to $\pi_1(X, x_0)/H$ if $\tilde{X}$ is a normal covering. Hence for the universal cover $\tilde{X} \rightarrow X$ we have $G(\tilde{X}) = \pi_1(X)$.

Proposition 1.1.11. *If an action of a group G on a space Y satisfies: each $y \in Y$ has a neighborhood U such that all the images $g(U)$ for varying $g \in G$ are disjoint. Then*

- (a) *The quotient map $p : Y \rightarrow Y/G, p(y) = Gy$, is a normal covering space.*
- (b) *G is the group of deck transformations of this covering space $Y \rightarrow Y/G$ if Y is path-connected.*
- (c) *G is isomorphic to $\pi_1(Y/G)/p_*(\pi_1(Y))$ if Y is path-connected and locally path-connected.*

Theorem 1.1.12. *Here is some consequences about $K(G, 1)$:*

- (1) *S^1 is a $K(\mathbb{Z}, 1)$. Closed surfaces other than S^2 and $\mathbb{RP}^2$ are $K(G, 1)$'s. Then lens space $S^\infty/\mathbb{Z}_m$ is a $K(\mathbb{Z}_m, 1)$.*
- (2) *$K(G, 1) \times K(H, 1)$ is a $K(G \times H, 1)$.*
- (3) *For a closed connected subspace K of S^3 , $S^3 \setminus K$ is a $K(G, 1)$.*

1.2 Easy Result

Example 1.2.1. *We have*

$$\begin{aligned}\pi_1(\mathbb{R}^3 \setminus S^1) &\cong \mathbb{Z} \\ \pi_1(S^3 \setminus S^1) &\cong \mathbb{Z}\end{aligned}$$

since it deformation retracts to $S^2 \vee S^1$.

Example 1.2.2. *For a genus g surface Σ_g ,*

$$\pi_1(\Sigma_g) = \langle a_1, b_1, \dots, a_n, b_n \mid [a_1, b_1] \cdots [a_n, b_n] \rangle$$

Example 1.2.3. *For a nonorientable genus g surface N_g ,*

$$\pi_1(N_g) = \langle a_1, \dots, a_n \mid a_1^2 \cdots a_n^2 \rangle$$

Example 1.2.4. *For the Klein bottle K ,*

$$\pi_1(K) = \langle a, b \mid aba^{-1}b \rangle$$

1.3 Exercise

Exercise 1.3.1 ([1] Exercise 1.2.3). *Show that the complement of a finite set of points in $\mathbb{R}^n$ is simply-connected if $n \geq 3$.*

Proof. It deformation retracts to a finite wedge sum of S^{n-1} , hence simply-connected. □

Exercise 1.3.2 ([1] Exercise 1.2.4). *Let $X \subset \mathbb{R}^3$ be the union of n lines through the origin. Compute $\pi_1(\mathbb{R}^3 \setminus X)$.*

Proof. It deformation retracts to S^2 minus $2n$ points, which deformation retracts to the wedge sum of $2n - 1$ circles, hence

$$\pi_1(\mathbb{R}^3 \setminus X) \cong \mathbb{Z}^{2n-1}$$

□

Exercise 1.3.3 ([1] Exercise 1.2.7). *Let X be the quotient space of S^2 obtained by identifying the north and south poles to a single point. Put a cell complex structure on X and use this to compute $\pi_1(X)$.*

Proof. The cell complex structure is given by: a 0-cell which is the collapsing point of the north pole N and south pole S , two 1-cells a, b which are two arcs connecting N and S , two 2-cells which are hemispheres attaching by ab^{-1} . Hence

$$\pi_1(X) \cong \langle a, b \mid ab^{-1} \rangle \cong \mathbb{Z}$$

□

Example 1.3.4 ([1] Exercise 1.2.8). *Compute the fundamental group of the space obtained from two tori $S^1 \times S^1$ by identifying a circle $S^1 \times \{x_0\}$ in one torus with the corresponding circle $S^1 \times \{x_0\}$ in the other torus.*

Proof. Applying the Van Kampen theorem to two tori, we get

$$\pi_1(X) \cong \langle a, b, c, d \mid aba^{-1}b^{-1}, cdc^{-1}d^{-1}, ac^{-1} \rangle \cong \langle a, b, d \mid aba^{-1}b^{-1}, ada^{-1}d^{-1} \rangle \cong (\mathbb{Z} * \mathbb{Z}) \times \mathbb{Z}$$

□

Exercise 1.3.5 ([1] Exercise 1.2.11). *Suppose T_f is the mapping torus of $f : X \rightarrow X$, compute a representation for $\pi_1(T_f)$ when $X = S^1 \wedge S^1$ and $X = S^1 \times S^1$.*

Proof. T_f can be seen as attaching 2-cells to $X \vee S^1$ along $a\gamma a^{-1}f_*(\gamma)^{-1}$, where $a \in \pi_1(S^1)$ is the generator, $\gamma \in \pi_1(X)$. Hence for $X = S^1 \vee S^1$,

$$\pi_1(T_f) = \langle a, b, c \mid bab^{-1} = f_*(a), cac^{-1} = f_*(c) \rangle$$

For $X = S^1 \times S^1$,

$$\pi_1(T_f) = \langle a, b, c \mid bc = cb, bab^{-1} = f_*(a), cac^{-1} = f_*(c) \rangle$$

□

Exercise 1.3.6 ([1] Exercise 1.2.14). *Consider the quotient space of a cube I^3 obtained by identifying each square face with the opposite square face via the right-handed screw motion consisting of a translation by one unit in the direction perpendicular to the face combined with a one-quarter twist of the face about its center point. Show this quotient space X is a cell complex with two 0-cells, four 1-cells, three 2-cells, and one 3-cell. Using this structure, show that $\pi_1(X)$ is the quaternion group of order eight.*

Proof. The cell structure is easy to obtain. Denote the four cells by a, b, c, d (by a suitable choice of orientation), then $\pi_1(X)$ is generated by $ab^{-1}, ac^{-1}, ad^{-1}$. Thus

$$\begin{aligned} \pi_1(X) &= \langle ab^{-1}, ac^{-1}, ad^{-1} \mid ad^{-1}bc^{-1}, ab^{-1}cd^{-1}, ac^{-1}db^{-1} \rangle \\ &= \langle x, y, z \mid zx^{-1}y, xy^{-1}z, yz^{-1}x \rangle \\ &= \langle x, y, z \mid x = yz, y = zx, z = xy \rangle \\ &= \langle x, y, z \mid x^2 = y^2 = z^2 = xyz \rangle \end{aligned}$$

which is Q_8 .

□

Remark 1.3.7. *Be careful that $\pi_1(X)$ is not generated by a, b, c, d since they are not loops.*

Exercise 1.3.8 ([1] Exercise 1.2.17). *Show that $\pi_1(\mathbb{R}^2 \setminus \mathbb{Q}^2)$ is uncountable.*

Proof. Denote $X = \mathbb{R}^2 \setminus \mathbb{Q}^2$. Let $x_0 = (\sqrt{2}, 0) \in X$, we want to find a surjection

$$f : \pi_1(X, x_0) \longrightarrow \prod_{r \in \mathbb{R}} \mathbb{Z}$$

For each $r \in \mathbb{R}$, define a loop

$$\gamma_r : (\sqrt{2}, 0) \longrightarrow (\sqrt{2}, r) \longrightarrow (\sqrt{2} + r, r) \longrightarrow (\sqrt{2} + r, 0) \longrightarrow (\sqrt{2}, 0)$$

then $[\gamma_r] \in \pi_1(X, x_0)$. We need to show that for $r_1 < r_2$, $[\gamma_{r_1}] \neq [\gamma_{r_2}]$. Choose $q \in \mathbb{Q}$, such that $r_1 < q < r_2$ and consider $q_0 = (\sqrt{2} + q, q)$. Let

$$i : X \longrightarrow \mathbb{R}^2 \setminus \{q_0\}$$

be the natural inclusion, then it induces

$$i_* : \pi_1(X, x_0) \longrightarrow \pi_1(\mathbb{R}^2 \setminus \{q_0\}, x_0)$$

If $[\gamma_{r_1}] = [\gamma_{r_2}] \in \pi_1(X, x_0)$, then $[\gamma_{r_1}] = [\gamma_{r_2}] \in \pi_1(\mathbb{R}^2 \setminus \{q_0\}, x_0)$. But in fact $[\gamma_{r_1}] = 0$ and $[\gamma_{r_2}] \neq 0$, which is a contradiction. Thus f is a surjection. $\square$

Exercise 1.3.9 ([1] Exercise 1.2.19). *Show that the subspace of $\mathbb{R}^3$ that is the union of the spheres of radius $1/n$ and center $(1/n, 0, 0)$ for $n = 1, 2, \dots$ is simply-connected.*

Proof. Denote $X = \cup_n B_n$ where B_n is the sphere with radius $1/n$. For any $\gamma : [0, 1] \rightarrow X$ such that $\gamma(0) = \gamma(1) = (0, 0, 0)$. We need to show that γ is nullhomotopic.

Consider $\gamma_n = \gamma([0, 1]) \cap B_n$, then γ_n is a loop in B_n based at $(0, 0, 0)$. Since B_n is simply-connected, γ_n is nullhomotopic. It can be deformation retracts to the constant map. Now we define a homotopy as follows: in time $[1/2^n, 1/2^{n-1}]$, F performs deformation retract from γ_n to constant map. Then F gives a homotopy from γ to the constant map. $\square$

Exercise 1.3.10 ([1] Exercise 1.3.9). *Show that if a path-connected, locally path-connected space X has $\pi_1(X)$ finite, then every map $X \rightarrow S^1$ is nullhomotopic.*

Proof. Consider the universal covering $p : \mathbb{R} \rightarrow S^1$. For any $f : X \rightarrow S^1$, since $\pi_1(X)$ is finite but $\pi_1(S^1) = \mathbb{Z}$, we have $f_*(\pi_1(X)) = 0$. Thus by the lifting criterion 1.1.8, f can be lifted to $\tilde{f} : X \rightarrow \mathbb{R}$. Since $\mathbb{R}$ is simply-connected, $\tilde{f}$ is nullhomotopic. $\square$

Exercise 1.3.11 ([1] Exercise 1.3.15). *Let $p : \tilde{X} \rightarrow X$ be a simply-connected covering space of X and let $A \subset X$ be a path-connected, locally path-connected subspace, with $\tilde{A} \subset \tilde{X}$ a path-component of $p^{-1}(A)$. Show that $p : \tilde{A} \rightarrow A$ is the covering space corresponding to the kernel of the map $\pi_1(A) \rightarrow \pi_1(X)$.*

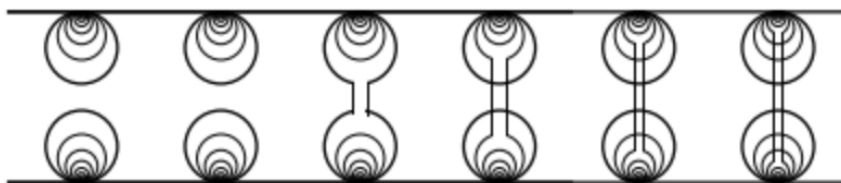
Proof. The point is to note that for $[\gamma] \in \pi_1(X, x_0)$, $[\gamma] = 0$ if and only if the lift of γ in $\tilde{X}$ is a loop. $\square$

Exercise 1.3.12 ([1] Exercise 1.3.6, [2] 26 spring T1). *Construct two covering maps $p : X \rightarrow Y, q : Y \rightarrow Z$ among topological spaces, such that $q \circ p$ is not a covering map.*

Proof. We take Z to be the Hawaii earring wedge S^1 , take Y to be



Then take X to be



It is easy to check the result.

□

Chapter 2

Homology

2.1 Basis

Method 2.1.1. To calculate homology, we can use *simplicial homology, singular homology(not useful), quotient, excision, cellular homology, MV sequence*.

Theorem 2.1.2. If (X, A) is a good pair, i.e. A is a nonempty closed subspace that is a deformation retract of some neighborhood in X , then there is an exact sequence

$$\cdots \rightarrow \tilde{H}_n(A) \xrightarrow{i_*} \tilde{H}_n(X) \xrightarrow{j_*} \tilde{H}_n(X/A) \xrightarrow{\partial} \tilde{H}_{n-1}(A) \rightarrow \cdots$$

Theorem 2.1.3 (Excision). Given $Z \subset A \subset X$ such that the closure of Z is contained in the interior of A , then the inclusion $(X - Z, A - Z) \hookrightarrow (X, A)$ induces isomorphisms $H_n(X - Z, A - Z) \rightarrow H_n(X, A)$ for all n . Equivalently, for subspaces A, B whose interiors cover X , the inclusion $(B, A \cap B) \hookrightarrow (X, A)$ induces isomorphisms $H_n(B, A \cap B) \rightarrow H_n(X, A)$ for all n .

Theorem 2.1.4. S^n has a continuous field of nonzero tangent vectors iff n is odd.

Theorem 2.1.5. $\mathbb{Z}_2$ is the only nontrivial group that can act freely on S^n if n is even.

Method 2.1.6. To calculate degree, we may use local degree.

Slogan 2.1.7. H_1 is the abelization of π_1 .

Theorem 2.1.8. An odd map $f : S^n \rightarrow S^n$ must have odd degree.

Theorem 2.1.9 (Lefschetz fixed point theorem). If X is a retract of a finite simplicial complex, and $f : X \rightarrow X$ is a map with $\tau(f) := \sum_n (-1)^n \text{tr}(f_* : H_n(X) \rightarrow H_n(X)) \neq 0$, then f has a fixed point.

2.2 Easy Result

Example 2.2.1. Homology of orientable genus g surface Σ_g :

$$H_2(\Sigma_g) = \mathbb{Z}, H_1(\Sigma_g) = \mathbb{Z}^{2g}, H_0(\Sigma_g) = \mathbb{Z}$$

Homology of nonorientable genus g surface N_g :

$$H_2(N_g) = \mathbb{Z}, H_1(N_g) = \mathbb{Z}^{g-1} \oplus \mathbb{Z}_2, H_0(N_g) = \mathbb{Z}$$

Homology of $\mathbb{CP}^n$:

$$H_i(\mathbb{CP}^n) = \begin{cases} \mathbb{Z}, & i = 0, 2, 4, \dots, 2n \\ 0, & \text{else} \end{cases}$$

Homology of $\mathbb{RP}^n$:

$$H_k(\mathbb{RP}^n) = \begin{cases} \mathbb{Z}, & k = 0 \text{ or } k = n \text{ odd} \\ \mathbb{Z}_2, & 0 < k < n \text{ and } k \text{ odd} \\ 0, & \text{else} \end{cases}$$

Homology of Lens space:

$$H_k(L_m(\ell_1, \dots, \ell_n)) = \begin{cases} \mathbb{Z}, & k = 0 \text{ or } k = 2n - 1 \\ \mathbb{Z}_m, & 0 < k < 2n - 1 \text{ and } k \text{ odd} \\ 0, & \text{else} \end{cases}$$

Example 2.2.2. Given $f, g : X \rightarrow Y$, form $Z = X \times I \sqcup Y / ((x, 0) \sim f(x), (x, 1) \sim g(x))$. Then there is an exact sequence

$$\dots \rightarrow H_n(X) \xrightarrow{f_* - g_*} H_n(Y) \xrightarrow{i_*} H_n(Z) \rightarrow H_{n-1}(X) \rightarrow \dots$$

Remark 2.2.3. When $f, g : X \rightarrow X$ and $g = \text{id}$, then $Z = T_f$ is called the **mapping torus** of f .

Example 2.2.4. Denote the suspension of X by SX , then we have

$$\tilde{H}_n(SX) \cong \tilde{H}_{n-1}(X)$$

Example 2.2.5. The homology of the connected sum of two manifolds M_1 and M_2 are given by:

$$H_i(M_1 \# M_2) \cong H_i(M_1) \oplus H_i(M_2), \quad 0 < i < n - 1$$

$$H_{n-1}(M_1 \# M_2) = \begin{cases} H_{n-1}(M_1) \oplus H_{n-1}(M_2), & \text{at least one is orientable} \\ H_{n-1}(M_1) \oplus H_{n-1}(M_2) & \text{by replacing one of the } \mathbb{Z}_2 \text{ summands by a } \mathbb{Z} \text{ summand} \end{cases}$$

2.3 Exercise

Exercise 2.3.1 ([1] Exercise 2.1.1). What familiar space is the quotient Δ -complex of a 2-simplex $[v_0, v_1, v_2]$ obtained by identifying the edges $[v_0, v_1]$ and $[v_1, v_2]$, preserving the ordering of vertices?

Proof. Möbius strip. □

Exercise 2.3.2 ([1] Exercise 2.1.4). Compute the simplicial homology groups of the triangular parachute obtained from Δ^2 by identifying its three vertices to a single point.

Proof. Using simplicial homology, we get

$$H_2(X) = 0, H_1(X) = \mathbb{Z}^2, H_0(X) = \mathbb{Z}$$

□

Exercise 2.3.3 ([1] Exercise 2.1.6). Compute the simplicial homology groups of the Δ -complex obtained from $n + 1$ 2-simplices $\Delta_0^2, \dots, \Delta_n^2$ by identifying all three edges of Δ_0^2 to a single edge, and for $i > 0$ identifying the edges $[v_0, v_1]$ and $[v_1, v_2]$ of Δ_i^2 to a single edge and the edge $[v_0, v_2]$ to the edge $[v_0, v_1]$ of Δ_{i-1}^2 .

Proof. Using simplicial homology, we get

$$H_2(X) = 0, H_1(X) = \mathbb{Z}_{2^n}, H_0(X) = \mathbb{Z}$$

□

Exercise 2.3.4 ([1] Exercise 2.1.9). Compute the homology groups of the Δ -complex X obtained from Δ^n by identifying all faces of the same dimension. Thus X has a single k -simplex for each $k \leq n$.

Proof. Using simplicial homology, we get

$$\begin{aligned} H_0(X) &= \mathbb{Z} \\ H_i(X) &= 0, \forall 1 \leq i \leq n-1 \\ H_n(X) &= \begin{cases} \mathbb{Z}, & n \text{ is odd} \\ 0, & n \text{ is even} \end{cases} \end{aligned}$$

□

Exercise 2.3.5 ([1] Exercise 2.1.11). *Show that if A is a retract of X then the map $H_n(A) \rightarrow H_n(X)$ induced by the inclusion $A \subset X$ is injective.*

Proof. Note that $r \circ i = \text{id}$, then by naturality.

□

Exercise 2.3.6 ([1] Exercise 2.1.16). (a) *Show that $H_0(X, A) = 0$ iff A meets each path-component of X .*

(b) *Show that $H_1(X, A) = 0$ iff $H_1(A) \rightarrow H_1(X)$ is surjective and each path-component of X contains at most one path-component of A .*

Proof. By long exact sequence of (X, A) , we immediately get the result.

□

Exercise 2.3.7. ([1] Exercise 2.1.17) *Compute the homology groups $H_n(X, A)$ when X is S^2 or $S^1 \times S^1$ and A is a finite set of points in X .*

Proof. By long exact sequence of (X, A) , we get

$$\begin{aligned} H_2(S^2, \cup_n *) &= \mathbb{Z}, H_1(S^2, \cup_n *) = \mathbb{Z}^{n-1}, H_0(S^2, \cup_n *) = 0 \\ H_2(S^1 \times S^1, \cup_n *) &= \mathbb{Z}, H_1(S^1 \times S^1, \cup_n *) = \mathbb{Z}^{n+1}, H_0(S^1 \times S^1, \cup_n *) = 0 \end{aligned}$$

□

Exercise 2.3.8 ([1] Exercise 2.1.18). *Show that for the subspace $\mathbb{Q} \subset \mathbb{R}$, the relative homology group $H_1(\mathbb{R}, \mathbb{Q})$ is free abelian and find a basis.*

Proof. By long exact sequence of $(\mathbb{R}, \mathbb{Q})$, we have

$$H_1(\mathbb{R}, \mathbb{Q}) \cong H_0(\mathbb{Q})$$

which is free abelian with countable basis $\{[0, q] \mid q \in \mathbb{Q}\}$.

□

Exercise 2.3.9 ([1] Exercise 2.1.19). *Compute the homology groups of the subspace of $I \times I$ consisting of the four boundary edges plus all points in the interior whose first coordinate is rational.*

Proof. Let $U = \{(x, y) \in X \mid y < 3/4\}$, $V = \{(x, y) \in X \mid y > 1/4\}$, then U, V are contractible. By MV sequence of $U \cup V = X$, we get

$$H_1(X) \cong H_0(U \cap V) = \mathbb{Z}^{\mathbb{N}}$$

and other homology groups are 0.

□

Exercise 2.3.10 ([1] Exercise 2.1.20). *Show that $\tilde{H}_n(X) \cong \tilde{H}_{n+1}(SX)$ for all n , where SX is the suspension of X . More generally, thinking of SX as the union of two cones CX with their bases identified, compute the reduced homology groups of the union of any finite number of cones CX with their bases identified.*

Proof. By the long exact sequence of (CX, X) and $CX/X \cong SX$, we get

$$\tilde{H}_n(X) \cong \tilde{H}_{n+1}(SX)$$

Note that $Y/CX \cong \bigvee_{n-1} SX$, we have

$$\tilde{H}_k(Y) \cong \bigoplus_{n-1} \tilde{H}_{k-1}(X)$$

□

Exercise 2.3.11 ([1] Exercise 2.1.26). *Show that $H_1(X, A)$ is not isomorphic to $\tilde{H}_1(X/A)$ if $X = [0, 1]$ and A is the sequence $1, \frac{1}{2}, \frac{1}{3}, \dots$ together with its limit 0.*

Proof. By long exact sequence of (X, A) , we have

$$H_1(X, A) \cong H_0(A) \cong \mathbb{Z}^{\mathbb{N}}$$

But we know that X/A is the Hawaii earring ring, and $\tilde{H}_1(X/A)$ is uncountable. Hence they are not isomorphic.

□

Exercise 2.3.12 ([1] Exercise 2.1.29). *Show that $S^1 \times S^1$ and $S^1 \vee S^1 \vee S^2$ have isomorphic homology groups in all dimensions, but their universal covering spaces do not.*

Proof. They have homology groups

$$H_2 \cong \mathbb{Z}, H_1 \cong \mathbb{Z}^2, H_0 \cong \mathbb{Z}$$

But the universal cover of $S^1 \times S^1$ is $\mathbb{R}^2$, hence contractible. The universal cover of $S^1 \vee S^1 \vee S^2$ contains infinitely many S^2 , hence it has nontrivial H_2 .

□

Exercise 2.3.13 ([1] Proposition 2.45). *If a finite-dimensional CW complex X is a $K(G, 1)$, then $G = \pi_1(X)$ must be torsionfree.*

Proof. If G has torsion, then it contains a subgroup $\mathbb{Z}_m$. Hence there exists a covering space Y of X such that $\pi_1(Y) = \mathbb{Z}_m$. Then Y is a $K(\mathbb{Z}_m, 1)$, which has the homotopy type of lens space $S^\infty/\mathbb{Z}_m$. Thus Y is infinite-dimensional, which is a contradiction.

□

Exercise 2.3.14 ([1] Exercise 2.2.1). *Prove the Brouwer fixed point theorem for maps $f : D^n \rightarrow D^n$ by applying degree theory to the map $S^n \rightarrow S^n$ that sends both the northern and southern hemispheres of S^n to the southern hemisphere via f .*

Proof. Consider a map $\phi : S^n \rightarrow S^n$ as follows: First map two hemispheres to the middle D^n , then use f mapping D^n to itself, then map D^n to the southern hemisphere.

On the one hand, ϕ maps through D^n , hence $\deg(\phi) = 0$. On the other hand, if f has no fixed points, then ϕ has no fixed points. Hence ϕ is homotopic to the antipodal map, which has degree $(-1)^{n+1} \neq 0$, a contradiction.

□

Exercise 2.3.15 ([1] Exercise 2.2.2). *Given a map $f : S^{2n} \rightarrow S^{2n}$, show that there is some point $x \in S^{2n}$ with either $f(x) = x$ or $f(x) = -x$. Deduce that every map $\mathbb{RP}^{2n} \rightarrow \mathbb{RP}^{2n}$ has a fixed point. Construct maps $\mathbb{RP}^{2n-1} \rightarrow \mathbb{RP}^{2n-1}$ without fixed points from linear transformations $\mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ without eigenvectors.*

Proof. If $f(x) \neq x$ for any x , then f is homotopic to the antipodal map. Hence $\deg f = (-1)^{2n+1} = -1$. If $f(x) \neq -x$ for any x , then f is homotopic to the identity map. Hence $\deg f = 1$, which is a contradiction.

For any map $f : \mathbb{RP}^{2n} \rightarrow \mathbb{RP}^{2n}$, we can compose with $\pi : S^{2n} \rightarrow \mathbb{RP}^{2n}$ to get $\bar{f} : S^{2n} \rightarrow \mathbb{RP}^{2n}$. By property of covering space, $\bar{f}$ can be lifted to $\tilde{f} : S^{2n} \rightarrow S^{2n}$. By the previous discussion, there exists $x \in S^{2n}$ such that

$\tilde{f}(x) = x$ or $\tilde{f}(x) = -x$. Hence $\pi(x)$ is a fixed point of f .

Define $f : \mathbb{RP}^{2n-1} \rightarrow \mathbb{RP}^{2n-1}$ by

$$f([x_1 : x_2 : \cdots : x_{2n}]) = [x_2 : x_1 : \cdots : x_{2n} : x_{2n-1}]$$

then f has no fixed points. □

Exercise 2.3.16 ([1] Exercise 2.2.3). *Let $f : S^n \rightarrow S^n$ be a map of degree zero. Show that there exist points $x, y \in S^n$ with $f(x) = x$ and $f(y) = -y$. Use this to show that if F is a continuous vector field defined on the unit ball D^n in $\mathbb{R}^n$ such that $F(x) \neq 0$ for all x , then there exists a point on ∂D^n where F points radially outward and another point on ∂D^n where F points radially inward.*

Proof. By the same argument of degree as last exercise, there exists x, y such that $f(x) = x, f(y) = -y$.

Define a map $f : \partial D^n \rightarrow D^n \rightarrow \partial D^n$ as follows: the first map is inclusion, the second map maps $x \in D^n$ to $F(x)/|F(x)|$. Since f factors through D^n , f has degree zero. By the previous discussion, there exists $x, y \in \partial D^n$ such that $f(x) = x, f(y) = -y$. Hence $F(x)$ points radially outwards, $F(y)$ points radially inwards. □

Exercise 2.3.17 ([1] Exercise 2.2.4). *Construct a surjective map $S^n \rightarrow S^n$ of degree zero, for each $n \geq 1$.*

Proof. First consider $n = 1$. Define $f : S^1 \rightarrow S^1$ to wrap S^1 once clockwise and once counterclockwise. Then f is surjective and $\deg f = 0$. Since $\deg Sf = \deg f$, where Sf is the suspension of f , for any n , we can take $S^{n-1}f$. □

Exercise 2.3.18 ([1] Exercise 2.2.6). *Show that every map $S^n \rightarrow S^n$ can be homotoped to have a fixed point if $n > 0$.*

Proof. Fix $x_0 \in S^n$ such that $f(x_0) \neq x_0$. Then there exists a rotation θ such that $\theta \cdot x_0 = f(x_0)$. Now we define a homotopy

$$\begin{aligned} F : S^n \times [0, 1] &\longrightarrow S^n \\ (x, t) &\longmapsto t\theta \cdot x \end{aligned}$$

Then $F(-, 1)$ has a fixed point $f(x_0)$. □

Exercise 2.3.19 ([1] Exercise 2.2.7). *For an invertible linear transformation $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ show that the induced map on $H_n(\mathbb{R}^n, \mathbb{R}^n - \{0\}) \cong \tilde{H}_{n-1}(\mathbb{R}^n - \{0\}) \cong \mathbb{Z}$ is id or $-\text{id}$ according to whether the determinant of f is positive or negative.*

Proof. f is represented by a matrix A . Note that doing elementary operations to A doesn't affect the homotopy class of f . Hence we may assume A is diagonal with diagonal entries $\{\pm 1\}$. Hence $f_* = \pm \text{id}$ depending on the number of -1 , which is $\det A$. □

Exercise 2.3.20 ([1] Exercise 2.2.8, [2] 26 spring T7). *A polynomial $f(z)$ with complex coefficients, viewed as a map $\mathbb{C} \rightarrow \mathbb{C}$, can always be extended to a continuous map of one-point compactifications $\hat{f} : S^2 \rightarrow S^2$. Show that the degree of $\hat{f}$ equals the degree of f as a polynomial. Show also that the local degree of $\hat{f}$ at a root of f is the multiplicity of the root.*

Proof. If $\deg f = 0$, then f is constant, hence

$$\deg f = 0 = \deg \hat{f}$$

Now we suppose $\deg f \geq 1$. Since $\deg \hat{f}$ is the sum of local degree, we may consider locally. Suppose z_0 is a root of f with multiplicity m , write

$$f(z) = (z - z_0)^m g(z)$$

where $g(z_0) \neq 0$, then the local degree $H^2(U_i, U_i - z_0) \rightarrow H^2(V, V - 0)$ is represented by a circle C around z_0 . Thus we only need to consider the degree of $\bar{f}$

$$\begin{aligned}\bar{f} : C &\longrightarrow S^1 \\ z &\longmapsto \frac{f(z)}{|f(z)|}\end{aligned}$$

Take C sufficiently small, then we can define a homotopy

$$\begin{aligned}F : S^1 \times [0, 1] &\longrightarrow S^1 \\ (z, t) &\longmapsto \frac{(z - z_0)^m (tg(z) + (1 - t)g(z_0))}{|(z - z_0)^m (tg(z) + (1 - t)g(z_0))|}\end{aligned}$$

Thus $\bar{f}$ is homotopic to $F(-, 0)$, which has degree m , thus the result holds. $\square$

Exercise 2.3.21 ([1] Exercise 2.2.9). *Compute the homology groups of the following 2-complexes:*

1. *The quotient of S^2 obtained by identifying north and south poles to a point.*
2. $S^1 \times (S^1 \vee S^1)$.
3. *The space obtained from D^2 by first deleting the interiors of two disjoint subdisks in the interior of D^2 and then identifying all three resulting boundary circles together via homeomorphisms preserving clockwise orientations of these circles.*
4. *The quotient space of $S^1 \times S^1$ obtained by identifying points in the circle $S^1 \times \{x_0\}$ that differ by $2\pi/m$ rotation and identifying points in the circle $\{x_0\} \times S^1$ that differ by $2\pi/n$ rotation.*

Proof. 1. Using the cellular homology, or note that X is homotopic to $S^2 \vee S^1$, we get

$$H_2(X) \cong \mathbb{Z}, H_1(X) \cong \mathbb{Z}, H_0(X) \cong \mathbb{Z}$$

2. Using the cellular homology or Künneth formula, we get

$$H_2(X) \cong \mathbb{Z}^2, H_1(X) \cong \mathbb{Z}^3, H_0(X) \cong \mathbb{Z}$$

3. Using the cellular homology, we get

$$H_2(X) = 0, H_1(X) \cong \mathbb{Z}^2, H_0(X) \cong \mathbb{Z}$$

4. Using the cellular homology, we get

$$H_2(X) \cong \mathbb{Z}, H_1(X) \cong \mathbb{Z}^2, H_0(X) \cong \mathbb{Z}$$

$\square$

Exercise 2.3.22 ([1] Exercise 2.2.10). *Let X be the quotient space of S^2 under the identifications $x \sim -x$ for x in the equator S^1 . Compute the homology groups $H_i(X)$. Do the same for S^3 with antipodal points of the equatorial $S^2 \subset S^3$ identified.*

Proof. Using the cellular homology, we get

$$H_2(X) \cong \mathbb{Z}, H_1(X) \cong \mathbb{Z}_2, H_0(X) \cong \mathbb{Z}$$

For $X = S^3 / \sim$, we have

$$H_3(X) \cong \mathbb{Z}^2, H_2(X) = 0, H_1(X) \cong \mathbb{Z}_2, H_0(X) \cong \mathbb{Z}$$

$\square$

Exercise 2.3.23 ([1] Exercise 2.2.11). In exercise 1.3.6 we described a 3-dimensional CW complex obtained from the cube I^3 by identifying opposite faces via a one-quarter twist. Compute the homology groups of this complex.

Proof. Using the cellular homology, we get

$$H_3(X) = \mathbb{Z}, H_2(X) = 0, H_1(X) \cong \mathbb{Z}_2^2, H_0(X) \cong \mathbb{Z}$$

□

Remark 2.3.24. $H_1(X)$ can also be verified by abelization of $\pi_1(X) \cong Q_8$.

Exercise 2.3.25 ([1] Exercise 2.2.12). Show that the quotient map $S^1 \times S^1 \rightarrow S^2$ collapsing the subspace $S^1 \vee S^1$ to a point is not nullhomotopic by showing that it induces an isomorphism on H_2 . On the other hand, show via covering spaces that any map $S^2 \rightarrow S^1 \times S^1$ is nullhomotopic.

Proof. The collapsing map preserves 2-cell and both $d_2 = 0$, hence it induces an isomorphism on H_2 , thus not nullhomotopic. By property of covering space, any map $f : S^2 \rightarrow S^1 \times S^1$ factors through $\mathbb{R}^2$, hence nullhomotopic. □

Exercise 2.3.26 ([1] Exercise 2.2.14). A map $f : S^n \rightarrow S^n$ satisfying $f(x) = f(-x)$ for all x is called an even map. Show that an even map $S^n \rightarrow S^n$ must have even degree, and that the degree must in fact be zero when n is even. When n is odd, show there exist even maps of any given even degree.

Proof. Suppose $f : S^n \rightarrow S^n$ is an even map, then f factors through $\mathbb{RP}^n$. If n is even, then $H_n(\mathbb{RP}^n) = 0$, hence $\deg f = 0$. If n is odd, say $f = g \circ \pi$, where $\pi : S^n \rightarrow \mathbb{RP}^n$, then

$$f_*[S^n] = g_*\pi_*[S^n] = 2g_*[\mathbb{RP}^n] = 2k[S^n]$$

hence $\deg f = 2k$ is even. Conversely, note that the quotient map $\mathbb{RP}^n \rightarrow \mathbb{RP}^n/\mathbb{RP}^{n-1} \simeq S^n$ induces an isomorphism on H_n , hence by composing with a degree k map $S^n \rightarrow S^n$, we can construct a map $g : \mathbb{RP}^n \rightarrow S^n$ such that

$$g_*[\mathbb{RP}^n] = k[S^n]$$

Hence $f = g \circ \pi$ satisfies $\deg f = 2k$. □

Example 2.3.27 ([1] Exercise 2.2.16). Let $\Delta^n = [v_0, \dots, v_n]$ have its natural Δ -complex structure with k -simplices $[v_{i_0}, \dots, v_{i_k}]$ for $i_0 < \dots < i_k$. Compute the ranks of the simplicial (or cellular) chain groups $\Delta_i(\Delta^n)$ and the subgroups of cycles and boundaries. Apply this to show that the k -skeleton of Δ^n has homology groups $\tilde{H}_i((\Delta^n)^k)$ equal to 0 for $i < k$, and free of rank $\binom{n}{k+1}$ for $i = k$.

Proof. Denote the rank of $\Delta_k(\Delta^n)$, by r_k , its cycles and boundaries by z_k, b_k . Then $r_k = \binom{n+1}{k+1}$. Since Δ^n is contractible, we have

$$H_0(\Delta^n) \cong \mathbb{Z}, H_i(\Delta^n) = 0, \forall i > 0$$

hence

$$z_0 - b_0 = 1, z_k = b_k, \forall k > 0$$

Moreover, we have

$$z_k + b_{k-1} = r_k = \binom{n+1}{k+1}$$

by isomorphism theorem. Hence we have

$$z_k = b_k = \binom{n}{k+1}, k > 0$$

Thus

$$\tilde{H}_k(\Delta^n)^k \cong \mathbb{Z}^{z_k} \cong \mathbb{Z}^{\binom{n}{k+1}}$$

□

Exercise 2.3.28 ([1] Exercise 2.2.19). Compute $H_i(\mathbb{RP}^n/\mathbb{RP}^m)$ for $m < n$ by cellular homology, using the standard CW structure on $\mathbb{RP}^n$ with $\mathbb{RP}^m$ as its m -skeleton.

Proof. Using cellular homology, we get

$$\begin{aligned} H_0(\mathbb{RP}^n/\mathbb{RP}^m) &\cong \mathbb{Z} \\ H_k(\mathbb{RP}^n/\mathbb{RP}^m) &\cong \mathbb{Z}, m+1 < k < n, k \text{ odd} \\ H_n(\mathbb{RP}^n/\mathbb{RP}^m) &\cong \mathbb{Z}, n \text{ odd} \\ H_{m+1}(\mathbb{RP}^n/\mathbb{RP}^m) &\cong \mathbb{Z}, m \text{ odd} \end{aligned}$$

□

Exercise 2.3.29 ([1] Exercise 2.2.20). For finite CW complexes X and Y , show that $\chi(X \times Y) = \chi(X)\chi(Y)$.

Proof. We have

$$\begin{aligned} \chi(X \times Y) &= \sum_k (-1)^k c_k(X \times Y) \\ &= \sum_k (-1)^k \sum_{i=1}^k c_i(X) \times c_{k-i}(Y) \\ &= \sum_k \sum_{i+j=k} (-1)^i c_i(X) \times (-1)^j c_j(Y) \\ &= \sum_i (-1)^i c_i(X) \times \sum_j (-1)^j c_j(Y) \\ &= \chi(X)\chi(Y) \end{aligned}$$

□

Exercise 2.3.30 ([1] Exercise 2.2.21). If a finite CW complex X is the union of subcomplexes A and B , show that $\chi(X) = \chi(A) + \chi(B) - \chi(A \cap B)$

Proof. We have

$$\begin{aligned} \chi(A \cup B) &= \sum_k (-1)^k c_k(A \cup B) \\ &= \sum_k (-1)^k (c_k(A) + c_k(B) - c_k(A \cap B)) \\ &= \chi(A) + \chi(B) - \chi(A \cap B) \end{aligned}$$

□

Exercise 2.3.31 ([1] Exercise 2.2.22). For X a finite CW complex and $p: \tilde{X} \rightarrow X$ an n -sheeted covering space, show that $\chi(\tilde{X}) = n\chi(X)$

Proof. By local homeomorphism of covering space, we immediately get the result. □

Exercise 2.3.32 ([1] Exercise 2.2.23). Show that if the closed orientable surface M_g of genus g is a covering space of M_h , then $g = n(h-1) + 1$ for some n , namely, n is the number of sheets in the covering.

Proof. Suppose M_g is a n -sheeted covering space of M_h , then by last exercise, $\chi(M_g) = n\chi(M_h)$, that is,

$$2 - 2g = n(2 - 2h)$$

Then we get the result. □

Exercise 2.3.33 ([1] Exercise 2.2.28). *Compute the homology groups of the following spaces:*

1. *The space obtained from a torus $S^1 \times S^1$ by attaching a Möbius band via a homeomorphism from the boundary circle of the Möbius band to the circle $S^1 \times \{x_0\}$ in the torus.*
2. *Do the same for the space obtained by attaching a Möbius band to $\mathbb{RP}^2$ via a homeomorphism of its boundary circle to the standard $\mathbb{RP}^1 \subset \mathbb{RP}^2$.*

Proof. 1. Using MV sequence, we get

$$H_2(X) \cong \mathbb{Z}, H_1(X) \cong \mathbb{Z}^2, H_0(X) \cong \mathbb{Z}$$

2. Using MV sequence, we get

$$H_2(X) = 0, H_1(X) \cong \mathbb{Z}_4, H_0(X) \cong \mathbb{Z}$$

□

Exercise 2.3.34 ([1] Exercise 2.2.29). *The surface M_g of genus g , embedded in $\mathbb{R}^3$ in the standard way, bounds a compact region R . Two copies of R , glued together by the identity map between their boundary surfaces M_g , form a closed 3-manifold X . Compute the homology groups of X and also compute the relative groups $H_i(R, M_g)$.*

Proof. Using MV sequence, we get

$$H_3(X) = \mathbb{Z}, H_2(X) \cong \mathbb{Z}^g, H_1(X) \cong \mathbb{Z}, H_0(X) \cong \mathbb{Z}$$

By long exact sequence of pair (R, M_g) , we get

$$H_3(R, M_g) \cong \mathbb{Z}, H_2(R, M_g) \cong \mathbb{Z}^g, H_1(R, M_g) = 0, H_0(R, M_g) \cong \mathbb{Z}$$

□

Exercise 2.3.35. *For the mapping torus T_f of a map $f : X \rightarrow X$, we have a long exact sequence $\cdots \rightarrow H_n(X) \xrightarrow{\mathbb{I}-f_*} H_n(X) \rightarrow H_n(T_f) \rightarrow H_{n-1}(X) \rightarrow \cdots$. Use this to compute the homology of the mapping tori of the following maps:*

1. *A reflection $S^2 \rightarrow S^2$.*
2. *A map $S^2 \rightarrow S^2$ of degree 2.*
3. *The map $S^1 \times S^1 \rightarrow S^1 \times S^1$ that is the identity on one factor and a reflection on the other.*
4. *The map $S^1 \times S^1 \rightarrow S^1 \times S^1$ that is a reflection on each factor.*
5. *The map $S^1 \times S^1 \rightarrow S^1 \times S^1$ that interchanges the two factors and then reflects one of the factors.*

Proof. 1. We have

$$H_3(T_f) = 0, H_2(T_f) \cong \mathbb{Z}_2, H_1(T_f) \cong \mathbb{Z}, H_0(T_f) \cong \mathbb{Z}$$

2. We have

$$H_3(T_f) = 0, H_2(T_f) = 0, H_1(T_f) \cong \mathbb{Z}, H_0(T_f) \cong \mathbb{Z}$$

3. We have

$$H_3(T_f) = 0, H_2(T_f) = \mathbb{Z} \oplus \mathbb{Z}_2, H_1(T_f) \cong \mathbb{Z}^2 \oplus \mathbb{Z}_2, H_0(T_f) \cong \mathbb{Z}$$

4. We have

$$H_3(T_f) \cong \mathbb{Z}, H_2(T_f) \cong \mathbb{Z}, H_1(T_f) \cong \mathbb{Z} \oplus \mathbb{Z}_2^2, H_0(T_f) \cong \mathbb{Z}$$

5. We have

$$H_3(T_f) \cong \mathbb{Z}, H_2(T_f) \cong \mathbb{Z}, H_1(T_f) \cong \mathbb{Z} \oplus \mathbb{Z}_2, H_0(T_f) \cong \mathbb{Z}$$

□

Exercise 2.3.36 ([1] Exercise 2.2.31). *Show that there are isomorphisms $\tilde{H}_n(X \vee Y) \cong \tilde{H}_n(X) \oplus \tilde{H}_n(Y)$ if the basepoints of X and Y that are identified in $X \vee Y$ are deformation retracts of neighborhoods $U \subset X$ and $V \subset Y$.*

Proof. It's easy to obtain the result by MV sequence. □

Exercise 2.3.37 ([1] Exercise 2.2.33). *Suppose the space X is the union of open sets $A_1, \dots, A_n$ such that each intersection $A_{i_1} \cap \dots \cap A_{i_k}$ is either empty or has trivial reduced homology groups. Show that $\tilde{H}_i(X) = 0$ for $i \geq n - 1$, and give an example showing this inequality is best possible, for each n .*

Proof. Using the MV sequence, and the conclusion follows from induction. Then we take S^{n-2} , which can be seen as the union of n faces of a $(n + 1)$ -simplex, which implies the inequality is sharp. □

Exercise 2.3.38 ([1] Exercise 2.2.36). *Show that $H_i(X \times S^n) \cong H_i(X) \oplus H_{i-n}(X)$ for all i and n , where $H_i = 0$ for $i < 0$ by definition.*

Proof. Since $H_*(S^n)$ is free, the result follows from the general Künneth formula. □

Exercise 2.3.39 ([2] 25 autumn T1). *Consider the topological space X given by the gluing of the following polygonal pieces:*

$$14\overline{6}, \ 521\overline{53}, \ \underline{23}4\overline{6}$$

Here the word $14\overline{6}$ mean the labeling scheme for the 3-gon is $e_1e_4e_6^{-1}$, and so for the other words. Each $e_i, 1 \leq i \leq 6$ is oriented counter-clock-wisely and the underline means the orientation of the edge is clock-wise. Compute the homotopy groups $\pi_i(X)$ and singular homology groups $H_i(X; \mathbb{Z})$ for all $i \geq 1$.

Proof. Using cellular homology, we get

$$H_i(X) = \begin{cases} \mathbb{Z}, & i = 0 \\ \mathbb{Z}^3 \oplus \mathbb{Z}_2, & i = 1 \\ 0, & i > 1 \end{cases}$$

By cellular structure, we get

$$\begin{aligned} \pi_1(X) &\cong \langle e_1, e_2, e_3, e_4, e_5, e_6 \mid e_1e_4e_6^{-1}, e_5e_2e_1^{-1}e_5e_3, e_2^{-1}e_3^{-1}e_4^{-1}e_6 \rangle \\ &\cong \langle a, b, c, d \mid daca^{-1}b^{-1}c^{-1}db \rangle \end{aligned}$$

Note that X is a closed surface, since it has infinite π_1 , X is a $K(\pi_1(X), 1)$. Thus $\pi_i(X) = 0$ for $i \geq 2$. □

Exercise 2.3.40. *Consider the quotient space $X = ([0, 1] \times S^1 \times S^1) / \sim$, where the equivalence relation $\sim$ is generated by*

$$(0, x, y) \sim (0, z, w) \quad \text{if} \quad xy = zw,$$

and

$$(1, x, y) \sim (1, z, w) \quad \text{if} \quad x^2y^6 = z^2w^6.$$

Here we treat S^1 as the space of unit complex numbers. Compute $H_n(X; \mathbb{Z})$ for all n .

Proof. Let $U = ([0, 1/2] \times S^1 \times S^1) / \sim$, $V = ([1/2, 1] \times S^1 \times S^1) / \sim$, then U, V both have the homotopy type S^1 . Moreover, $U \cap V \simeq S^1 \times S^1$. By MV sequence, we have a long exact sequence

$$0 \rightarrow H_3(X) \rightarrow H_2(U \cap V) \rightarrow H_2(U) \oplus H_2(V) \rightarrow H_2(X) \rightarrow H_1(U \cap V) \xrightarrow{i_*} H_1(U) \oplus H_1(V) \rightarrow H_1(X) \rightarrow 0$$

We know that $H_2(U \cap V) \cong \mathbb{Z}$, hence $H_3(X) \cong \mathbb{Z}$. Note that $i_* : \mathbb{Z}^2 \rightarrow \mathbb{Z}^2$ is represented by the matrix $\begin{pmatrix} 1 & 1 \\ 2 & 6 \end{pmatrix}$ thus $H_2(X) = 0$, $H_1(X) \cong \mathbb{Z}_4$. Since X is connected, $H_0(X) \cong \mathbb{Z}$. And other homology groups are 0. $\square$

Exercise 2.3.41 ([1] Proposition 2.29, [3] 24 SF T5). *Suppose n is even. Prove that the only nontrivial group that acts freely on S^n is $\mathbb{Z}_2$.*

Proof. For any $g \in G$, g defines a map $g : S^n \rightarrow S^n$ which has no fixed points. Hence g is homotopic to the antipodal map. Thus $\deg(g) = (-1)^{2n+1} = -1$. But note that

$$\deg(gh) = \deg(g) \deg(h) = 1$$

Hence G has at most one element excluding identity. Thus G can only be $\mathbb{Z}_2$. And $\mathbb{Z}_2$ indeed acts freely on S^n by antipodal map. $\square$

Chapter 3

Cohomology

3.1 Basis

Theorem 3.1.1 (Universal coefficient theorem for cohomology). *If a chain complex C of free abelian groups has homology groups $H_n(C)$, then the cohomology groups $H^n(C; G)$ of the cochain complex $\text{Hom}(C_n, G)$ are determined by split exact sequences*

$$0 \longrightarrow \text{Ext}(H_{n-1}(C), G) \longrightarrow H^n(C; G) \longrightarrow \text{Hom}(H_n(C), G) \longrightarrow 0$$

Theorem 3.1.2 (Universal coefficient theorem for homology). *If C is a chain complex of free abelian groups, then there are natural short exact sequences*

$$0 \rightarrow H_n(C) \otimes G \rightarrow H_n(C; G) \rightarrow \text{Tor}(H_{n-1}(C), G) \rightarrow 0$$

and these sequences split.

Method 3.1.3. *We can use the following to compute $\text{Ext}(H, G)$ for finitely generated H :*

1. $\text{Ext}(H \oplus H', G) \cong \text{Ext}(H, G) \oplus \text{Ext}(H', G)$.
2. $\text{Ext}(H, G) = 0$ if H is free.
3. $\text{Ext}(\mathbb{Z}_n, G) \cong G/nG$.

Slogan 3.1.4. *The exact sequence is natural.*

Theorem 3.1.5. *There is a long exact sequence*

$$\cdots \longrightarrow H^n(X, A; G) \xrightarrow{j^*} H^n(X; G) \xrightarrow{i^*} H^n(A; G) \xrightarrow{\delta} H^{n+1}(X, A; G) \longrightarrow \cdots$$

Theorem 3.1.6 (MV sequence). *There is a long exact sequence for $X = \text{int}A \cap \text{int}B$:*

$$\cdots \longrightarrow H^n(X; G) \xrightarrow{\Psi} H^n(A; G) \oplus H^n(B; G) \xrightarrow{\Phi} H^n(A \cap B; G) \longrightarrow H^{n+1}(X; G) \longrightarrow \cdots$$

Theorem 3.1.7. $\alpha \smile \beta = (-1)^{|\alpha||\beta|} \beta \smile \alpha$ holds for all $\alpha \in H^k(X, A; R)$ and $\beta \in H^l(X, A; R)$ when R is commutative.

Theorem 3.1.8 (Künneth formula). *The cross product $H^*(X; R) \otimes_R H^*(Y; R) \rightarrow H^*(X \times Y; R)$ is an isomorphism of rings if X and Y are CW complexes and $H^k(Y; R)$ is a finitely generated free R -module for all k .*

Theorem 3.1.9. *Cohomology of the Grassmannian (classifying space $BU(n)$) is $H^*(\text{Gr}_n(\mathbb{C}^\infty); \mathbb{Z}) \cong \mathbb{Z}[\alpha_1, \dots, \alpha_n]$ with $|\alpha_i| = 2i$. Cohomology of the classifying space $BSU(n)$ is $H^*(BSU(n); \mathbb{Z}) \cong \mathbb{Z}[\alpha_2, \dots, \alpha_n]$ with $|\alpha_i| = 2i$.*

Theorem 3.1.10. *Let M be a closed connected n -manifold. Then:*

1. *If M is R -orientable, the map $H_n(M; R) \rightarrow H_n(M|x; R) \cong R$ is an isomorphism for all $x \in M$.*
2. *If M is not R -orientable, the map $H_n(M; R) \rightarrow H_n(M|x; R) \cong R$ is injective with image $\{r \in R | 2r = 0\}$ for all $x \in M$.*
3. *$H_i(M; R) = 0$ for $i > n$.*

Proposition 3.1.11. *If M is a closed connected n -manifold, the torsion subgroup of $H_{n-1}(M; \mathbb{Z})$ is trivial if M is orientable and $\mathbb{Z}_2$ if M is nonorientable.*

Proposition 3.1.12. *If M is a connected noncompact n -manifold, then $H_i(M; R) = 0$ for $i \geq n$.*

Theorem 3.1.13 (Poincaré duality). *If M is a closed R -orientable n -manifold, then the map $D : H^k(M; R) \rightarrow H_{n-k}(M; R)$ defined by $D(\alpha) = [M] \frown \alpha$ is an isomorphism for all k .*

Method 3.1.14. *To calculate cohomology with compact support, we may use*

$$H_c^i(X; G) = \varinjlim_K H^i(X, X \setminus K; G)$$

Theorem 3.1.15 (Poincaré duality for noncompact manifold). *Suppose M is an R -oriented n -manifold, then $D_M : H_c^k(M; R) \rightarrow H_{n-k}(M; R)$ is an isomorphism for all k .*

Proposition 3.1.16. *A closed manifold of odd dimension has Euler characteristic zero.*

Theorem 3.1.17 (Lefschetz duality). *Suppose M is a compact R -orientable n -manifold whose boundary ∂M is decomposed as the union of two compact $(n-1)$ -dimensional manifolds A and B with a common boundary $\partial A = \partial B = A \cap B$. Then cap product with a fundamental class $[M] \in H_n(M, \partial; R)$ gives isomorphisms $D_M : H^k(M, A; R) \rightarrow H_{n-k}(M, B; R)$ for all k .*

Theorem 3.1.18 (Alexander duality). *If K is a compact, locally contractible, nonempty, proper subspace of S^n , then $\tilde{H}_i(S^n \setminus K; \mathbb{Z}) \cong \tilde{H}^{n-i-1}(K; \mathbb{Z})$ for all i .*

Slogan 3.1.19. *Cup product is transverse intersection of submanifolds:*

$$[M]^* \smile [N]^* = [M \frown N]^*$$

Theorem 3.1.20 (General Künneth formula). *If X and Y are CW complexes and R is a principal ideal domain, then there are natural short exact sequences*

$$0 \rightarrow \bigoplus_i (H_i(X; R) \otimes_R H_{n-i}(Y; R)) \rightarrow H_n(X \times Y; R) \rightarrow \bigoplus_I \text{Tor}(H_i(X; R), H_{n-i-1}(Y; R)) \rightarrow 0$$

and these sequences split.

3.2 Easy Result

Example 3.2.1. *Cohomology of orientable genus g surface Σ_g :*

$$\begin{aligned} H^2(\Sigma_g) &= \mathbb{Z} = \mathbb{Z}\{T\} \\ H^1(\Sigma_g) &= \mathbb{Z}^{2g} = \mathbb{Z}\{\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n\} \\ H^0(\Sigma_g) &= \mathbb{Z} \end{aligned}$$

the ring structure is given by

$$\alpha_i \smile \beta_j = \delta_{ij} T, \quad \alpha_i \smile \alpha_j = 0, \quad \beta_i \smile \beta_j = 0$$

Example 3.2.2. *$H^*(\mathbb{RP}^n; \mathbb{Z}_2) \cong \mathbb{Z}_2[\alpha]/(\alpha^{n+1})$ and $H^*(\mathbb{RP}^\infty; \mathbb{Z}_2) \cong \mathbb{Z}_2[\alpha]$ where $|\alpha| = 1$. In the complex case, $H^*(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}[\alpha]/(\alpha^{n+1})$ and $H^*(\mathbb{CP}^\infty; \mathbb{Z}) \cong \mathbb{Z}[\alpha]$ where $|\alpha| = 2$*

3.3 Exercise

Exercise 3.3.1 ([1] Exercise 3.1.9). *Show that if $f : S^n \rightarrow S^n$ has degree d then $f^* : H^n(S^n; G) \rightarrow H^n(S^n; G)$ is multiplication by d .*

Proof. This follows from the naturality of universal coefficient theorem. $\square$

Exercise 3.3.2 ([1] Exercise 3.1.10). *For the lens space $L_m(\ell_1, \dots, \ell_n)$, compute the cohomology groups using the cellular cochain complex and taking coefficients in $\mathbb{Z}, \mathbb{Q}, \mathbb{Z}_m$, and $\mathbb{Z}_p$ for p prime.*

Proof. We have

$$\begin{aligned} H^k(L_m(\ell_1, \dots, \ell_n); \mathbb{Z}) &= \begin{cases} \mathbb{Z}, & k = 0, 2n-1 \\ \mathbb{Z}_m, & 0 < k < 2n-1, k \text{ even} \\ 0, & \text{else} \end{cases} \\ H^k(L_m(\ell_1, \dots, \ell_n); \mathbb{Q}) &= \begin{cases} \mathbb{Q}, & k = 0, 2n-1 \\ 0, & \text{else} \end{cases} \\ H^k(L_m(\ell_1, \dots, \ell_n); \mathbb{Z}_m) &= \begin{cases} \mathbb{Z}_m, & 0 \leq k \leq 2n-1 \\ 0, & \text{else} \end{cases} \\ H^k(L_m(\ell_1, \dots, \ell_n); \mathbb{Z}_p) &= \begin{cases} \mathbb{Z}_p, & k = 0, 2n-1 \\ \mathbb{Z}_{\gcd(p, m)}, & 0 < k < 2n-1 \\ 0, & \text{else} \end{cases} \end{aligned}$$

$\square$

Exercise 3.3.3 ([1] Exercise 3.2.2). *Using the cup product $H^k(X, A; R) \times H^\ell(X, B; R) \rightarrow H^{k+\ell}(X, A \cup B; R)$, show that if X is the union of contractible open subsets A and B , then all cup products of positive-dimensional classes in $H^*(X; R)$ are zero. This applies in particular if X is a suspension. Generalize to the situation that X is the union of n contractible open subsets, to show that all n -fold cup products of positive-dimensional classes are zero.*

Proof. By long exact sequence of pair (X, A) , and by $\tilde{H}^*(A) = 0$, we have

$$\tilde{H}^k(X) \cong H^k(X, A)$$

and since $A \cup B = X$, we have $H^*(X, A \cup B) = 0$. Thus the cup product is trivial. For n open subsets, we consider

$$H^{k_1}(X, A_1; R) \times \dots \times H^{k_n}(X, A_n; R) \rightarrow H^{k_1 + \dots + k_n}(X, A_1 \cup \dots \cup A_n; R)$$

and the argument is the same. $\square$

Exercise 3.3.4 ([1] Exercise 3.2.3). 1. *Using the cup product structure, show there is no map $\mathbb{RP}^n \rightarrow \mathbb{RP}^m$ inducing a nontrivial map $H^1(\mathbb{RP}^m; \mathbb{Z}_2) \rightarrow H^1(\mathbb{RP}^n; \mathbb{Z}_2)$ if $n > m$. What is the corresponding result for maps $\mathbb{CP}^n \rightarrow \mathbb{CP}^m$?*

2. *Prove the Borsuk-Ulam theorem by the following argument. Suppose on the contrary that $f : S^n \rightarrow \mathbb{R}^n$ satisfies $f(x) \neq f(-x)$ for all x . Then define $g : S^n \rightarrow S^{n-1}$ by $g(x) = (f(x) - f(-x))/|f(x) - f(-x)|$, so $g(-x) = -g(x)$ and g induces a map $\mathbb{RP}^n \rightarrow \mathbb{RP}^{n-1}$. Show that part (a) applies to this map.*

Proof. 1. Suppose $f : \mathbb{RP}^n \rightarrow \mathbb{RP}^m$, then f induces $f^* : H^*(\mathbb{RP}^m; \mathbb{Z}_2) \rightarrow H^*(\mathbb{RP}^n; \mathbb{Z}_2)$. Suppose $H^*(\mathbb{RP}^m; \mathbb{Z}_2) = \mathbb{Z}_2[\alpha]/(\alpha^{m+1})$, $H^*(\mathbb{RP}^n; \mathbb{Z}_2) = \mathbb{Z}_2[\beta]/(\beta^{n+1})$, then say $f^*(\alpha) = \beta$. Thus we have

$$0 = f^*(\alpha^{m+1}) = (f^*(\alpha))^{m+1} = \beta^{m+1}$$

Hence $m \geq n$. Similarly, there is no map $\mathbb{CP}^n \rightarrow \mathbb{CP}^m$ if $n > m$.

2. Follows from description. □

Exercise 3.3.5 ([1] Exercise 3.2.4, Exercise 2C.3, [3] 25 SF T6). *Show that every map $f : \mathbb{CP}^n \rightarrow \mathbb{CP}^n$ has a fixed point if n is even. When n is odd show there is a fixed point unless $f^*(\alpha) = -\alpha$, for α a generator of $H^2(\mathbb{CP}^n; \mathbb{Z})$, and give an example that there are no fixed points.*

Proof. f induces a ring homomorphism $f^* : H^*(\mathbb{CP}^n; \mathbb{Z}) \rightarrow H^*(\mathbb{CP}^n; \mathbb{Z})$. Suppose $\alpha \in H^2(\mathbb{CP}^n; \mathbb{Z})$ is a generator, then we can write

$$f^*(\alpha) = k\alpha$$

for some integer k . By ring homomorphism,

$$f^*(\alpha^i) = k^i \alpha^i$$

By the universal coefficient theorem, the homology homomorphism induced by f :

$$f_* : H_{2i}(\mathbb{CP}^n; \mathbb{Z}) \rightarrow H_{2i}(\mathbb{CP}^n; \mathbb{Z})$$

is multiplication by k^i . Hence the Lefschetz number of f is

$$\tau(f) = \sum_{i=0}^n (-1)^{2i} k^i = \begin{cases} \frac{1-k^{n+1}}{1-k}, & \text{if } k \neq 1 \\ n+1, & \text{if } k = 1 \end{cases}$$

Hence only the case that $k = -1$ and n is odd can make $\tau(f) = 0$. Thus for other cases, by Lefschetz fixed point theorem, f has a fixed point.

Consider an example $f : \mathbb{CP}^{2n-1} \rightarrow \mathbb{CP}^{2n-1}$ defined by

$$f([z_1, \dots, z_{2n}]) = [\bar{z}_2, -\bar{z}_1, \dots, \bar{z}_{2n}, -\bar{z}_{2n-1}]$$

then f doesn't have a fixed point. □

Example 3.3.6 ([1] Exercise 3.2.5). *Show the ring $H^*(\mathbb{RP}^\infty; \mathbb{Z}_{2k})$ is isomorphic to $\mathbb{Z}_{2k}[\alpha, \beta]/(2\alpha, 2\beta, \alpha^2 - k\beta)$ where $|\alpha| = 1$ and $|\beta| = 2$.*

Proof. By cellular cohomology, we obtain

$$H^k(\mathbb{RP}^\infty; \mathbb{Z}_{2k}) \cong \mathbb{Z}_2, \forall k > 0$$

If we denote the generator of H^1 by α , generator of H^2 by β , and the generator of $H^1(\mathbb{RP}^\infty; \mathbb{Z}_2)$ by γ . Then we have $2\alpha = 0, 2\beta = 0$, and $k\alpha$ corresponds to $k\gamma$, and β corresponds to γ^2 . Thus we obtain a relation $\alpha^2 - k^2\beta = 0$. By dimension argument, we have

$$H^*(\mathbb{RP}^\infty; \mathbb{Z}_{2k}) \cong \mathbb{Z}_{2k}[\alpha, \beta]/(2\alpha, 2\beta, \alpha^2 - k^2\beta)$$

We can check this is the same as in the exercise. □

Exercise 3.3.7 ([1] Exercise 3.2.6). *Compute the map $H^*(\mathbb{CP}^n; \mathbb{Z}) \rightarrow H^*(\mathbb{CP}^n; \mathbb{Z})$ induced by the map $\mathbb{CP}^n \rightarrow \mathbb{CP}^n$ that is a quotient of the map $\mathbb{C}^{n+1} \rightarrow \mathbb{C}^{n+1}$ raising each coordinate to the d^{th} power, $(z_0, \dots, z_n) \mapsto (z_0^d, \dots, z_n^d)$, for a fixed integer $d > 0$.*

Proof. We only need to calculate $H^2(\mathbb{CP}^n; \mathbb{Z}) \rightarrow H^2(\mathbb{CP}^n; \mathbb{Z})$, then by cup product we get the general consequence. And note that the inclusion $\mathbb{CP}^1 \hookrightarrow \mathbb{CP}^n$ induces an isomorphism on H^2 , thus we only need to consider $H^2(\mathbb{CP}^1; \mathbb{Z}) \rightarrow H^2(\mathbb{CP}^1; \mathbb{Z})$. Moreover, $\mathbb{CP}^1 \simeq S^2$, and by universal coefficient theorem, it is equivalent to consider $H_2(S^2; \mathbb{Z}) \rightarrow H_2(S^2; \mathbb{Z})$, which is the degree of the map. Thus we can use local degree to calculate it, and obtain it is multiplication by d . Thus

$$f^* : H^{2k}(\mathbb{CP}^n; \mathbb{Z}) \rightarrow H^{2k}(\mathbb{CP}^n; \mathbb{Z})$$

is multiplication by d^k . □

Exercise 3.3.8 ([1] Exercise 3.2.7). *Show that $\mathbb{RP}^3$ is not homotopy equivalent to $\mathbb{RP}^2 \vee S^3$.*

Proof. Suppose α is a generator for $H^1(\mathbb{RP}^3; \mathbb{Z}_2)$, then $\alpha^3 \neq 0$. But if β is a generator for $H^1(\mathbb{RP}^2 \vee S^3; \mathbb{Z}_2) \cong H^1(\mathbb{RP}^2; \mathbb{Z}_2)$, then $\beta^3 = 0$. They have different cohomology rings, hence are not homotopy equivalent. □

Exercise 3.3.9 ([1] Exercise 3.2.8). *Let X be $\mathbb{CP}^2$ with a cell e^3 attached by a map $S^2 \rightarrow \mathbb{CP}^1 \subset \mathbb{CP}^2$ of degree p , and let $Y = M(\mathbb{Z}_p, 2) \vee S^4$. Thus X and Y have the same 3-skeleton but differ in the way their 4-cells are attached. Show that X and Y have isomorphic cohomology rings with $\mathbb{Z}$ coefficients but not with $\mathbb{Z}_p$ coefficients.*

Proof. Direct calculation shows that both spaces have

$$H^4(; \mathbb{Z}) \cong \mathbb{Z}, H^3(; \mathbb{Z}) \cong \mathbb{Z}_p, H^2(; \mathbb{Z}) = 0$$

Hence all cup products are trivial. But for $\mathbb{Z}_p$,

$$H^4(; \mathbb{Z}_p) \cong \mathbb{Z}_p, H^3(; \mathbb{Z}_p) \cong \mathbb{Z}_p, H^2(; \mathbb{Z}_p) \cong \mathbb{Z}_p$$

The cup product $H^2(X; \mathbb{Z}_p) \smile H^2(X; \mathbb{Z}_p)$ is nontrivial because it comes from the ring structure of $\mathbb{CP}^2$. However, the ring structure of $H^*(Y; \mathbb{Z}_p)$ is trivial. □

Exercise 3.3.10 ([1] Exercise 3.2.11). *Show that every map $S^{k+\ell} \rightarrow S^k \times S^\ell$ induces the trivial homomorphism $H_{k+\ell}(S^{k+\ell}) \rightarrow H_{k+\ell}(S^k \times S^\ell)$, assuming $k > 0$ and $\ell > 0$.*

Proof. Since the cup product of $H^*(S^k \times S^\ell)$ is trivial, the induced map $f^* : H^{k+\ell}(S^k \times S^\ell) \rightarrow H^{k+\ell}(S^{k+\ell})$ is trivial. By universal coefficient theorem, $f_* : H_{k+\ell}(S^{k+\ell}) \rightarrow H_{k+\ell}(S^k \times S^\ell)$ is trivial. □

Exercise 3.3.11 ([1] Exercise 3.3.7). *Show that for any connected closed orientable n -manifold M there is a degree 1 map $M \rightarrow S^n$.*

Proof. Fix $x_0 \in M$, take a small neighborhood U around x_0 , then consider the contraction map which maps $M \setminus U$ to a single point. It is a degree ± 1 map by local degree. □

Exercise 3.3.12 ([1] Exercise 3.3.10). *Show that for a degree 1 map $f : M \rightarrow N$ of connected closed orientable manifolds, the induced map $f_* : \pi_1 M \rightarrow \pi_1 N$ is surjective, hence also $f_* : H_1(M) \rightarrow H_1(N)$.*

Proof. Consider the covering space $p : \tilde{N} \rightarrow N$ such that $p_* \pi_1(\tilde{N}) = f_* \pi_1(M)$. Hence f can be lifted to $\tilde{f} : M \rightarrow \tilde{N}$. Since M, N are closed, $\tilde{N}$ is also closed, thus p is a finite cover. Now we have

$$1 = \deg f = \deg(p \circ \tilde{f}) = \deg(p) \deg(\tilde{f})$$

Hence $\deg(p) = \pm 1$, which implies that $\tilde{N} = N$. Thus $f_* \pi_1(M) = \pi_1(N)$, that is, f_* is surjective. Homology comes from abelization of π_1 . □

Exercise 3.3.13 ([1] Exercise 3.3.11). *If M_g denotes the closed orientable surface of genus g , show that degree 1 maps $M_g \rightarrow M_h$ exist iff $g \geq h$.*

Proof. If there exists a degree 1 map, then by last exercise we know that $g \geq h$. Conversely, if $g \geq h$, we can define a contraction map by collapsing $g - h$ genus to a single point. It is a degree 1 map by local degree. $\square$

Exercise 3.3.14 ([1] Exercise 3.3.12). *Show that in a free group F with basis $x_1, \dots, x_{2k}$, the product of commutators $[x_1, x_2] \cdots [x_{2k-1}, x_{2k}]$ is not equal to a product of fewer than k commutators $[v_i, w_i]$ of elements $v_i, w_i \in F$.*

Proof. Suppose $[x_1, x_2] \cdots [x_{2k-1}, x_{2k}] = [v_1, w_1] \cdots [v_j, w_j]$. Then by CW complex, we can construct a degree map $f : M_j \rightarrow M_k$, which is a contradiction with last exercise. $\square$

Exercise 3.3.15 ([1] Exercise 3.3.13). *Let $M'_h \subset M_g$ be a compact subsurface of genus h with one boundary circle, so M'_h is homeomorphic to M_h with an open disk removed. Show there is no retraction $M_g \rightarrow M'_h$ if $h > g/2$.*

Proof. Suppose there exists a retraction $r : M_g \rightarrow M'_h$, then $r^* : H^*(M'_h) \rightarrow H^*(M_g)$ is an injection. Note that the Poincaré pairing for M_g

$$\mathbb{Z}^{2g} \times \mathbb{Z}^{2g} \rightarrow \mathbb{Z}$$

is nonsingular, but the Poincaré pairing for M'_h is always degenerate. Hence $2h < g$, which implies $h < g/2$. $\square$

Exercise 3.3.16 ([1] Exercise 3.3.24, [2] 26 spring T3, [2] 25 spring T7). *Let M be a closed connected 3-manifold, and write $H_1(M; \mathbb{Z})$ as $\mathbb{Z}^r \oplus F$, the direct sum of a free abelian group of rank r and a finite group F . Show that $H_2(M; \mathbb{Z})$ is $\mathbb{Z}^r$ if M is orientable and $\mathbb{Z}^{r-1} \oplus \mathbb{Z}_2$ if M is nonorientable. In particular, $r \geq 1$ when M is nonorientable. Construct examples showing there are no other restrictions on the homology groups of closed 3-manifolds.*

Proof. By 3.1.11 and note that $\chi(M) = 0$, we obtain the form of H_2 . For M orientable, we have $H_1(S^1 \times S^2) \cong \mathbb{Z}$, $H_1(L_m) \cong \mathbb{Z}_m$ for lens space L_m . Hence by taking connected sum, we get any orientable 3-manifold with prescribed H_1 . For M nonorientable, consider the manifold N obtained from $S^2 \times I$ by identifying $S^2 \times \{0\}$ with $S^2 \times \{1\}$ via a reflection of S^2 . Then N is nonorientable and $H_1(N) \cong \mathbb{Z}$. By taking connected sum of N and lens spaces, we get any nonorientable 3-manifold with prescribed H_1 . $\square$

Exercise 3.3.17 ([1] Exercise 3.3.25). *Show that if a closed orientable manifold M of dimension $2k$ has $H_{k-1}(M; \mathbb{Z})$ torsionfree, then $H_k(M; \mathbb{Z})$ is also torsionfree.*

Proof. By Poincaré duality, $H_k(M; \mathbb{Z}) \cong H^k(M; \mathbb{Z})$. By universal coefficient theorem,

$$H^k(M; \mathbb{Z}) \cong \text{Hom}(H_k(M; \mathbb{Z}), \mathbb{Z}) \oplus \text{Ext}(H_{k-1}(M; \mathbb{Z}), \mathbb{Z})$$

Since $H_{k-1}(M; \mathbb{Z})$ is torsionfree, the Ext term vanishes. Hence $H_k(M; \mathbb{Z}) \cong H^k(M; \mathbb{Z})$ must be torsionfree. $\square$

Exercise 3.3.18 ([2] 26 spring T5). *Let $S \subset \mathbb{R}^3$ be a smoothly embedded closed surface. Prove that S is orientable.*

Proof. $S \subset \mathbb{R}^3 \subset S^3$. By Alexander duality, we have

$$H^2(S; \mathbb{Z}) \cong \tilde{H}_0(S^3 \setminus S; \mathbb{Z})$$

which is free abelian. But if S is nonorientable, $H_1(S; \mathbb{Z})$ contains a $\mathbb{Z}_2$ summand, hence by universal coefficient theorem, $H^2(S; \mathbb{Z})$ contains a

$$\text{Ext}(\mathbb{Z}_2, \mathbb{Z}) = \mathbb{Z}_2$$

summand, which is a contradiction! Thus S is orientable. $\square$

Exercise 3.3.19 ([2] 25 autumn T5). *Suppose X is contractible, compact manifold of dimension $n \geq 1$. Prove that $H_i(\partial X; \mathbb{Z}) = H_i(S^{n-1}; \mathbb{Z})$ for all i .*

Proof. Consider the long exact sequence of pair $(X, \partial X)$:

$$\cdots \longrightarrow \tilde{H}_i(\partial X) \longrightarrow \tilde{H}_i(X) \longrightarrow H_i(\partial X, X) \longrightarrow \tilde{H}_{i-1}(\partial X) \longrightarrow \cdots$$

Since X is contractible, $\tilde{H}_i(X) = 0$ for all i . Thus

$$\tilde{H}_i(\partial X) \cong H_{i+1}(\partial X, X)$$

Since X is contractible, it is orientable. By Lefschetz duality 3.1.17, we have

$$H_{i+1}(\partial X, X) \cong H^{n-i-1}(X)$$

which is $\mathbb{Z}$ for $i = n - 1$, and 0 otherwise. Hence

$$\tilde{H}_i(\partial X) = \begin{cases} \mathbb{Z}, & i = 0, n - 1 \\ 0, & \text{else} \end{cases}$$

which is just $H_i(S^{n-1}; \mathbb{Z})$. □

Exercise 3.3.20 ([1] Exercise 3.3.26, [2] 25 spring T5). *Compute the cohomology ring of $H^*((S^2 \times S^8) \# (S^4 \times S^6); \mathbb{Z})$.*

Proof. By Künneth formula and cohomology for connected sum, we immediately get

$$H^*((S^2 \times S^8) \# (S^4 \times S^6); \mathbb{Z}) \cong \mathbb{Z}[\alpha, \beta, \gamma, \delta] / (\alpha^2, \beta^2, \gamma^2, \delta^2, \alpha\gamma, \alpha\delta, \beta\gamma, \beta\delta, \alpha\beta - \gamma\delta)$$

where $|\alpha| = 2, |\beta| = 8, |\gamma| = 4, |\delta| = 6$. □

Exercise 3.3.21 ([1] Exercise 3.C.5). *Recall that a space is called an **H-space** if there exists a point $e \in X$ and a continuous map $\mu : X \times X \rightarrow X$ such that the map $X \rightarrow X$ defined by $x \mapsto \mu(e, x)$ and the map $X \rightarrow X$ defined by $x \mapsto \mu(x, e)$ are both homotopic to the identity map. Show that $\pi_1(X, e)$ is abelian.*

Proof. Consider

$$\pi_1(X, e) \rightarrow \pi_1(X \times X, e \times e) = \pi_1(X, e) \oplus \pi_1(X, e) \xrightarrow{\mu_*} \pi_1(X, e)$$

mapping $\alpha \in \pi_1(X, e)$ to $(\alpha, 0) \in \pi_1(X \times X, e \times e)$ then to $\mu(\alpha, e)$. By the homotopy in the definition of a H-space, we know that $\mu_*(\alpha, 0) = \alpha$. Similarly, if we let the first right arrow above to be the inclusion to the second coordinate, then we get $\mu_*(0, \alpha) = \alpha$.

Thus for any $\alpha, \beta \in \pi_1(X, e)$, we have

$$\alpha\beta = \mu_*(0, \alpha)\mu_*(\beta, 0) = \mu_*(\alpha, \beta) = \mu_*(\beta, 0)\mu_*(0, \alpha) = \beta\alpha$$

hence $\pi_1(X)$ is abelian. □

Exercise 3.3.22 ([2] 24 autumn T5). *Show that $\mathbb{CP}^n$ is not an H-space.*

Proof. For $\mathbb{CP}^n$, suppose it is an H-space, consider the action by μ on the cohomology ring of $\mathbb{CP}^n$:

$$H^*(\mathbb{CP}^n) \xrightarrow{\mu^*} H^*(\mathbb{CP}^n \times \mathbb{CP}^n)$$

We have known that $H^*(\mathbb{CP}^n) \cong \mathbb{Z}[x]/(x^{n+1})$ and by the Künneth formula $H^*(\mathbb{CP}^n \times \mathbb{CP}^n) \cong H^*(\mathbb{CP}^n) \otimes H^*(\mathbb{CP}^n) = \mathbb{Z}[x_1, x_2]/(x_1^{n+1}, x_2^{n+1})$. If we compose μ^* with the induced homomorphism by the inclusion of first coordinate i_1^* :

$$H^*(\mathbb{CP}^n) \xrightarrow{\mu^*} H^*(\mathbb{CP}^n \times \mathbb{CP}^n) \xrightarrow{i_1^*} H^*(\mathbb{CP}^n)$$

then it is an isomorphism by the homotopy to identity. The same argument holds for i_2^* , thus we conclude that $\mu^*(x) = x_1 + x_2$. However, $0 = \mu^*(x^{n+1}) = (x_1 + x_2)^{n+1} \neq 0$, which is a contradiction. □

Exercise 3.3.23. If $f : M \rightarrow N$ is a degree one map between compact connected oriented manifolds without boundary, show that the induced map on π_1 is surjective. Show that the induced map on $H_k(-)$ is surjective.

Proof. $f : M \rightarrow N$ induces $f_* : \pi_1(M) \rightarrow \pi_1(N)$. Let $p : \tilde{N} \rightarrow N$ be a covering map such that $p_*(\pi_1(\tilde{N})) = f_*(\pi_1(M))$, then there exists $\tilde{f} : M \rightarrow \tilde{N}$ making the following diagram commute by the lifting property.

$$\begin{array}{ccc} & & \tilde{N} \\ & \nearrow \tilde{f} & \downarrow p \\ M & \xrightarrow{f} & N \end{array}$$

If $[\pi_1(N), f_*(\pi_1(M))] = k$, $1 < k < \infty$, then $\deg P = k > 1$, but

$$1 = \deg f = \deg P \cdot \deg \tilde{f}$$

which is a contradiction. If $[\pi_1(N), f_*(\pi_1(M))] = \infty$, then $\tilde{N}$ is non-compact, f induces an isomorphism $f_* : H_n(M) \rightarrow H_n(N)$ and factors through $\tilde{N}$. However, $H_n(\tilde{N}) = 0$, which is a contradiction. Hence the only possible case is $[\pi_1(N), f_*(\pi_1(M))] = 1$, that is, f_* is surjective.

As for the homology groups, by the Poincaré duality, we have an isomorphism given by

$$\begin{aligned} H^{n-k}(N) &\xrightarrow{\cong} H_k(N) \\ \alpha &\mapsto \alpha \frown [N] \end{aligned}$$

Thus given an element $a = \alpha \frown [N]$ for some $\alpha \in H^{n-k}(N)$ in $H_k(N)$, let $b = f^*(\alpha) \frown [M]$, then

$$f_*(b) = f_*(f^*(\alpha) \frown [M]) = \alpha \frown f_*([M]) = \alpha \frown [N] = b$$

since $\deg f = 1$. □

Exercise 3.3.24 ([2] 24 autumn T7). Let M be an oriented, connected, closed manifold of dimension ≥ 2 . Let $f : S^n \rightarrow M$ be a continuous map of degree 1. Show that f must be a homotopy equivalence.

Proof. By last exercise, M is simply-connected, and $\tilde{H}_k(M) = 0$ for $k < n$. Hence by proposition 4.1.14, f is a homotopy equivalence. □

Exercise 3.3.25 ([3] 24 SF T1). Let $n > 1$ be a positive integer.

(1) Does there exist a map $f : S^{2n} \rightarrow \mathbb{CP}^n$ with $\deg(f) \neq 0$? Construct an example or disprove it.

(2) Does there exist a map $f : \mathbb{CP}^n \rightarrow S^{2n}$ with $\deg(f) \neq 0$? Construct an example or disprove it.

Proof. (1) Doesn't exist. For any $f : S^{2n} \rightarrow \mathbb{CP}^n$, f induces $f^* : H^*(\mathbb{CP}^n; \mathbb{Z}) \rightarrow H^*(S^{2n}; \mathbb{Z})$. Suppose α is a generator of $H^2(\mathbb{CP}^n) \cong \mathbb{Z}$. Since $n > 1$, $H^2(S^{2n}) = 0$. Thus $f^*(\alpha) = 0$. Hence $f^*(\alpha^n) = 0$, where α is a generator of $H^{2n}(\mathbb{CP}^n) \cong \mathbb{Z}$. By universal coefficient theorem, we have a commutative diagram

$$\begin{array}{ccc} H^{2n}(\mathbb{CP}^n; \mathbb{Z}) & \xrightarrow{\cong} & \text{Hom}(H_{2n}(\mathbb{CP}^n; \mathbb{Z})) \\ f^* \uparrow & & f^* \uparrow \\ H^{2n}(S^{2n}; \mathbb{Z}) & \xrightarrow{\cong} & \text{Hom}(H_{2n}(S^{2n}; \mathbb{Z})) \end{array}$$

Thus $f_* : H_{2n}(S^{2n}) \rightarrow H_{2n}(\mathbb{CP}^n)$ is zero. That is, $\deg(f) = 0$.

(2) Exist. Fix a point $p \in \mathbb{CP}^n$ and a small neighborhood $U \simeq D^{2n}$ of p . Define f by retracting on $\mathbb{CP}^n \setminus U$ to a single point of S^{2n} . Then f has local degree ± 1 . Hence $\deg(f) = \pm 1$. □

Exercise 3.3.26 ([3] 24 SF T3). *Let M be a closed, simply connected 6-dimensional manifold. Suppose $H_2(M) = \mathbb{Z}_2$. Prove that the Euler characteristic $\chi(M) \neq -1$.*

Proof. Since M is simply-connected, M is orientable. Hence $H_1(M) = 0, H_6(M) \cong \mathbb{Z}$. Since $H_2(M) \cong \mathbb{Z}_2$, by universal coefficient theorem, $H^1(M) = H^2(M) = 0$. Then by Poincaré duality, $H_4(M) = H_5(M) = 0$. By Poincaré duality, the pairing

$$\begin{aligned} H_3(M) \times H_3(M) &\longrightarrow \mathbb{Z} \\ (\alpha, \beta) &\longmapsto (\alpha^* \smile \beta^*)[M] \end{aligned}$$

is nondegenerate. Note that this pairing is anti-linear, hence $\text{rank}(H_3(M))$ is even. Thus

$$\chi(M) = \sum_{k=0}^6 (-1)^k \text{rank}(H_k(M)) = 2 - \text{rank}(H_3(M))$$

is even. Hence $\chi(M) \neq -1$. □

Chapter 4

Homotopy theory

4.1 Basis

Proposition 4.1.1. *A covering space projection $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ induces isomorphisms $p_* : \pi_n(\tilde{X}, \tilde{x}_0) \rightarrow \pi_n(X, x_0)$ for $n \geq 2$.*

Theorem 4.1.2. *There is a long exact sequence*

$$\cdots \rightarrow \pi_n(A, x_0) \xrightarrow{i_*} \pi_n(X, X_0) \xrightarrow{j_*} \pi_n(X, A, x_0) \xrightarrow{\partial} \pi_{n-1}(A, x_0) \rightarrow \cdots \rightarrow \pi_0(X, x_0)$$

Theorem 4.1.3 (Whitehead theorem). *If a map $f : X \rightarrow Y$ between connected CW complexes induces isomorphisms $f_* : \pi_n(X) \rightarrow \pi_n(Y)$ for all n , then f is a homotopy equivalence. In case f is an inclusion, we can obtain deformation retract.*

Theorem 4.1.4 (Cellular approximation theorem). *Every map $f : X \rightarrow Y$ of CW complexes is homotopic to a cellular map.*

Proposition 4.1.5. *Every space X has a CW approximation $f : Z \rightarrow X$.*

Proposition 4.1.6. *For every pair (X, A) with A a nonempty CW complex there exist n -connected CW models $f : (Z, A) \rightarrow (X, A)$ for all $n \geq 0$, and is unique up to homotopy equivalence rel A .*

Proposition 4.1.7. *A weak homotopy equivalence $f : X \rightarrow Y$ induces isomorphisms $f_* : H_n(X; G) \rightarrow H_n(Y; G)$ and $f^* : H^n(Y; G) \rightarrow H^n(X; G)$ for all n and G .*

Proposition 4.1.8. *A weak homotopy equivalence $f : Y \rightarrow Z$ induces bijections $[X, Y] \rightarrow [X, Z]$ and $\langle X, Y \rangle \rightarrow \langle X, Z \rangle$ for all X .*

Lemma 4.1.9 (Compression lemma). *$(X, A), (Y, B)$ CW pairs. For each n such that $X \setminus A$ has cells of dimension n , assume that $\pi_n(Y, B, y_0) = 0$ for all $y_0 \in B$. Then every map $f : (X, A) \rightarrow (Y, B)$ is homotopic rel A to a map $X \rightarrow B$.*

Lemma 4.1.10 (Extension lemma). *Given a CW pair (X, A) and a map $f : A \rightarrow Y$ with Y path-connected, then f can be extended to a map $x \rightarrow Y$ if $\pi_{n-1}(Y) = 0$ for all n such that $X \setminus A$ has cells of dimension n .*

Theorem 4.1.11. *Let X be a CW complex decomposed as the union of subcomplexes A and B with nonempty connected intersection $C = A \cap B$. If (A, C) is m -connected and (B, C) is n -connected, $m, n \geq 0$, then the map $\pi_i(A, C) \rightarrow \pi_i(X, B)$ induced by inclusion is an isomorphism for $i < m + n$ and a surjection for $i = m + n$.*

Theorem 4.1.12 (Freudenthal suspension theorem). *The suspension map $\pi_i(S^n) \rightarrow \pi_{i+1}(S^{n+1})$ is an isomorphism for $i < 2n - 1$ and a surjection for $i = 2n - 1$. More generally this holds for the suspension $\pi_i(X) \rightarrow \pi_{i+1}(SX)$ whenever X is an $(n - 1)$ -connected CW complex.*

Theorem 4.1.13 (Hurewicz theorem). *If a space X is $(n-1)$ -connected, $n \geq 2$, then $\tilde{H}_i(X) = 0$ for $i < n$ and $\pi_n(X) \cong H_n(X)$. If a pair (X, A) is $(n-1)$ -connected, $n \geq 2$, with A simply-connected and nonempty, then $H_i(X, A) = 0$ for $i < n$ and $\pi_n(X, A) \cong H_n(X, A)$.*

Proposition 4.1.14. *A map $f : X \rightarrow Y$ between simply-connected CW complexes is a homotopy equivalence if $f_* : H_n(X) \rightarrow H_n(Y)$ is an isomorphism for all n .*

Theorem 4.1.15. *Suppose $p : E \rightarrow B$ has the homotopy lifting property with respect to disks D^k for all $k \geq 0$. Choose basepoints $b_0 \in B$ and $x_0 \in F = p^{-1}(b_0)$. Then the map $p_* : \pi_n(E, F, x_0) \rightarrow \pi_n(B, b_0)$ is an isomorphism for all $n \geq 1$. Hence if B is path-connected, there is a long exact sequence*

$$\cdots \rightarrow \pi_n(F, x_0) \rightarrow \pi_n(E, x_0) \xrightarrow{p_*} \pi_n(B, b_0) \rightarrow \pi_{n-1}(F, x_0) \rightarrow \cdots \rightarrow \pi_0(E, x_0) \rightarrow 0.$$

This holds in particular for a fiber bundle.

Theorem 4.1.16. *There are natural bijections $T : \langle X, K(G, n) \rangle \rightarrow H^n(X; G)$ for all CW complexes X and all $n > 0$, with G any abelian group. Such a T has the form $T([f]) = f^*(\alpha)$ for a certain distinguished class $\alpha \in H^n(K(G, n); G)$.*

Proposition 4.1.17. *If X is a connected abelian CW complex and (W, A) is a CW pair such that $H^{n+1}(W, A; \pi_n X) = 0$ for all n , then every map $A \rightarrow X$ can be extended to a map $W \rightarrow X$.*

4.2 Easy Result

Example 4.2.1 (Postnikov tower). *For each connected CW complex X and n , there exists a CW complex X_n containing X as a subcomplex such that:*

(a) $\pi_i(X_n) = 0$ for $i > n$.

(b) The inclusion $X \hookrightarrow X_n$ induces an isomorphism on π_i for $i \leq n$.

Example 4.2.2 (Whitehead tower). *For any CW complex X , there exists Z_n*

$$\cdots \rightarrow Z_2 \rightarrow Z_1 \rightarrow Z_0 \rightarrow X$$

with Z_n n -connected and the map $Z_n \rightarrow X$ inducing an isomorphism on all π_i with $i > n$.

Example 4.2.3. $\pi_n(S^n) \cong \mathbb{Z}$, which can be given by the degree map.

Example 4.2.4. $\mathbb{CP}^\infty$ is a $K(\mathbb{Z}, 2)$, $\mathbb{RP}^\infty$ is a $K(\mathbb{Z}_2, 1)$.

Example 4.2.5. By spectral sequence, we have $\pi_4(S^2) \cong \mathbb{Z}_2$.

Example 4.2.6. We have the following two important fiber bundles:

$$O(n-1) \rightarrow O(n) \rightarrow S^{n-1}$$

$$U(n-1) \rightarrow U(n) \rightarrow S^{2n-1}$$

Thus we can obtain

$$\pi_1(O(n), x_0) = \pi_1(SO(n)) = \begin{cases} 0, n = 1 \\ \mathbb{Z}, n = 2 \\ \mathbb{Z}_2, n \geq 3 \end{cases}$$

$$\pi_2(O(n), x_0) = \pi_2(SO(n)) = 0$$

$$\pi_1(U(n)) \cong \mathbb{Z}$$

$$\pi_2(U(n)) = 0$$

$$\pi_3(U(n)) = \begin{cases} 0, n = 1 \\ \mathbb{Z}, n \geq 2 \end{cases}$$

$$\pi_1(SU(n)) = 0$$

$$\pi_2(SU(n)) = 0$$

$$\pi_3(SU(n)) = \begin{cases} 0, n = 1 \\ \mathbb{Z}, n \geq 2 \end{cases}$$

Moreover, note that $O(n)$ is the deformation retraction of $GL(n, \mathbb{R})$, $SO(n)$ is the deformation retraction of $GL(n, \mathbb{R})$, $U(n)$ is the deformation retraction of $GL(n, \mathbb{C})$, $SU(n)$ is the deformation retraction of $SL(n, \mathbb{C})$, we can compute many homotopy groups of matrix groups.

4.3 Exercise

Exercise 4.3.1 ([1] Exercise 4.1.11). *Show that a CW complex is contractible if it is the union of an increasing sequence of subcomplexes $X_1 \subset X_2 \subset \dots$ such that each inclusion $X_i \hookrightarrow X_{i+1}$ is nullhomotopic. An example is S^∞ , or more generally the infinite suspension $S^\infty X$ of any CW complex X .*

Proof. First we prove that $\pi_n(X) = 0$ for all n . Choose an element $[\alpha] \in \pi_n(X)$ where $\alpha : S^n \rightarrow X$. By cellular approximation theorem 4.1.4, we may assume $\alpha : S^n \rightarrow X^n$. Since $X^n \hookrightarrow X^{n+1}$ is nullhomotopic, hence $[\alpha] = 0$. Now consider $f : X \rightarrow \{*\}$, then f is a weak homotopy equivalence, hence by Whitehead theorem 4.1.3, X is contractible. $\square$

Exercise 4.3.2 ([1] Exercise 4.1.12). *Show that an n -connected, n -dimensional CW complex is contractible.*

Proof. Consider the identity map $(X, x_0) \rightarrow (X, x_0)$, then for any m -cells of X , $m \leq n$, and $\pi_m(X, x_0) = 0$. By compression lemma 4.1.9, then identity map can be homotoped to the constant map. Hence X is contractible. $\square$

Exercise 4.3.3 ([1] Exercise 4.1.13). *Show that a CW complex retracts onto any contractible subcomplex.*

Proof. Suppose A is a contractible subcomplex of X . Consider the identity map $A \rightarrow A$, since $\pi_i(A) = 0$, by extension lemma 4.1.10, it can be extended to $r : X \rightarrow A$. Then r is a retraction. $\square$

Exercise 4.3.4 ([1] Exercise 4.1.14). *Show that the n -skeletons of homotopy equivalent CW complexes without cells of dimension $n + 1$ are also homotopy equivalent.*

Proof. Suppose $F : X \rightarrow Y, G : Y \rightarrow X$ gives an homotopy equivalence between X, Y . Then by cellular approximation theorem 4.1.4, we may assume $F(X^n) \subset Y^n, G(Y^n) \subset X^n$. Suppose $K : X \times I \rightarrow X$ gives a homotopy between $G \circ F$ and id_X . By cellular approximation theorem 4.1.4 again, we may assume $K(X^n \times I) \subset X^{n+1} = X^n$, hence $K|_{X^n \times I}$ also gives a homotopy between $G|_{X^n} \circ F|_{X^n}$. Thus $F_{X^n} : X \rightarrow Y, G_{Y^n} : Y \rightarrow X$ gives an homotopy equivalence between X^n, Y^n . $\square$

Exercise 4.3.5 ([1] Exercise 4.1.16). *Show that a map $f : X \rightarrow Y$ between connected CW complexes factors as a composition $X \rightarrow Z_n \rightarrow Y$ where the first map induces isomorphisms on π_i for $i \leq n$ and the second map induces isomorphisms on π_i for $i \geq n + 1$.*

Proof. First take $Z_n = X$, and let $X \rightarrow Z_n$ be the identity map and $Z_n \rightarrow Y$ just be f . Then attaching $(n + 1)$ -cells for every generators of $\pi_{n+1}(Y)$ via constant maps and extend f to $Z_n \rightarrow Y$. This implies that $X \rightarrow Z_n$ induces isomorphisms on π_i for $i \leq n$ and $Z_n \rightarrow Y$ induces a surjection on π_{n+1} . Then attaching $(n + 2)$ -cells to Z_n to eliminate relations of $\pi_{n+1}(Y)$ and add generators of $\pi_{n+2}(Y)$. This implies that $Z_n \rightarrow Y$ induces an isomorphism on π_{n+1} and a surjection on π_{n+2} . Continuing this process, we get required Z_n . $\square$

Exercise 4.3.6 ([1] Exercise 4.1.17). *Show that if X and Y are CW complexes with X m -connected and Y n -connected, then $(X \times Y, X \vee Y)$ is $(m + n + 1)$ -connected, as is the smash product $X \wedge Y$.*

Proof. Since X is m -connected, X^m can be homotoped to a single point by the compression lemma 4.1.9. Thus we may assume X doesn't have cells of dimension 1 to m . Similarly, we may assume Y doesn't have cells of dimension 1 to n . Hence $X \vee Y$ is the $(m + n + 1)$ -skeleton of $X \times Y$. Thus $(X \times Y, X \vee Y)$ is $(m + n + 1)$ -connected. $\square$

Exercise 4.3.7 ([1] Exercise 3.1.18). *Give an example of a weak homotopy equivalence $X \rightarrow Y$ for which there does not exist a weak homotopy equivalence $Y \rightarrow X$.*

Proof. Take X to be a two points space, Y is the topologist's sine curve. Let $f(!) = (0, 0)$, $f(?) = (1, \sin 1)$. Then f is a weak homotopy equivalence. However, any continuous map $f : Y \rightarrow X$ must be constant, hence there doesn't exist a weak homotopy equivalence $Y \rightarrow X$. $\square$

Exercise 4.3.8 ([1] Exercise 4.2.1). *Show that there is no retraction $\mathbb{RP}^n \rightarrow \mathbb{RP}^k$ if $n > k > 0$.*

Proof. If there exists a retraction $r : \mathbb{RP}^n \rightarrow \mathbb{RP}^k$, then $r_* : \pi_k(\mathbb{RP}^n) \rightarrow \pi_k(\mathbb{RP}^k)$ is a surjection. But $\pi_k(\mathbb{RP}^n) = \pi_k(S^n) = 0$, $\pi_k(\mathbb{RP}^k) = \pi_k(S^k) \cong \mathbb{Z}$, which is a contradiction. $\square$

Exercise 4.3.9 ([1] Exercise 4.2.8). *Show that the suspension of an acyclic CW complex is contractible.*

Proof. Suppose X is acyclic. By Van Kampen's theorem, $\pi_1(SX) = 0$. By example 2.2.4,

$$\tilde{H}_n(SX) \cong \tilde{H}_{n-1}(X) = 0$$

Hence by Hurewicz theorem 4.1.13, $\pi_i(SX) = 0$ for all i . Then by Whitehead theorem 4.1.3, SX is contractible. $\square$

Exercise 4.3.10 ([1] Exercise 4.2.9). *Show that a map between simply-connected CW complexes is a homotopy equivalence if its mapping cone is contractible. Give an example where this fails in the nonsimply-connected case.*

Proof. Suppose $f : X \rightarrow Y$ where X, Y are simply-connected, then we have a long exact sequence

$$\cdots H_n(X) \xrightarrow{f_*} H_n(Y) \rightarrow H_n(Cf) \rightarrow H_{n+1}(X) \rightarrow \cdots$$

Since Cf is contractible, $f_* : H_n(X) \rightarrow H_n(Y)$ is an isomorphism for all i . Then by proposition 4.1.14, f is a homotopy equivalence.

To give a counterexample, we consider X to be a acyclic space which is not contractible (for example, the punctured Poincaré homology ball), then SX is contractible by last exercise. Let f be the inclusion $X \hookrightarrow CX$, then $Cf = SX$ is contractible, but f is not a homotopy equivalence. $\square$

Exercise 4.3.11 ([1] Exercise 4.2.12). *Show that a map $f : X \rightarrow Y$ of connected CW complexes is a homotopy equivalence if it induces an isomorphism on π_1 and a lift $\tilde{f} : \tilde{X} \rightarrow \tilde{Y}$ to the universal covers induces an isomorphism on homology.*

Proof. Since $\tilde{X}, \tilde{Y}$ are simply-connected, by proposition 4.1.14, $\tilde{f}$ induces an isomorphism on homotopy group. But we know that

$$\pi_n(X) \cong \pi_n(\tilde{X}), \pi_n(Y) \cong \pi_n(\tilde{Y}), n \geq 2$$

and that f induces an isomorphism on π_1 , hence f induces isomorphisms on all homotopy groups. By Whitehead theorem 4.1.3, f is a homotopy equivalence. $\square$

Exercise 4.3.12 ([1] Exercise 4.2.13). *Show that a map between connected n -dimensional CW complexes is a homotopy equivalence if it induces an isomorphism on π_i for $i \leq n$.*

Proof. We pass $f : X \rightarrow Y$ to $\tilde{f} : \tilde{X} \rightarrow \tilde{Y}$, where $\tilde{X}, \tilde{Y}$ are universal covers of X, Y . Then $\tilde{f}$ induces an isomorphism on π_i for $i \leq n$. We may assume X is an inclusion of Y (otherwise we may replace Y with M_f), then $\pi_i(\tilde{Y}, \tilde{X}) = 0$ for $i \leq n$ by long exact sequence. By Hurewicz theorem 4.1.13, $H_i(\tilde{Y}, \tilde{X}) = 0$ for all $i \leq n$.

Since $\tilde{X}, \tilde{Y}$ are n -dimensional, $H_i(\tilde{Y}, \tilde{X}) = 0$ for all $i > n$ by definition. Thus $\tilde{f} : \tilde{X} \rightarrow \tilde{Y}$ induces an isomorphism on homology. By last exercise, f is a homotopy equivalence. $\square$

Exercise 4.3.13 ([1] Exercise 4.2.14). *If an n -dimensional CW complex X contains a subcomplex Y homotopy equivalent to S^n , show that the map $\pi_n(Y) \rightarrow \pi_n(X)$ induced by inclusion is injective.*

Proof. Consider the Hurewicz homomorphism

$$\begin{array}{ccc} \pi_n(Y) & \xrightarrow{i_*} & \pi_n(X) \\ \downarrow & & \downarrow \\ H_n(Y) & \xrightarrow{i_*} & H_n(X) \end{array}$$

Since Y is homotopy equivalent to S^n , $\pi_n(Y) \cong \mathbb{Z}$, $H_n(Y) \cong \mathbb{Z}$ and $\pi_n(Y) \rightarrow H_n(Y)$ is an isomorphism. By cellular homology, $i_* : H_n(Y) \rightarrow H_n(X)$ is injective, hence $\pi_n(Y) \rightarrow \pi_n(X)$ is injective. $\square$

Exercise 4.3.14 ([1] Exercise 4.2.15). *Show that a closed simply-connected 3-manifold is homotopy equivalent to S^3 .*

Proof. We know that $H_3(M) \cong \mathbb{Z}$. Since $\pi_1(M) = 0$, $H_1(M) = 0$. By universal coefficient theorem, $H^1(M) = 0$. By Poincaré duality, $H_2(M) \cong H^1(M) = 0$. Now consider a map $f : M \rightarrow S^3$ such that $f_*[M] = [S^3]$, then f induces an isomorphism on homology between simply-connected spaces, hence by proposition 4.1.14, f is a homotopy equivalence. $\square$

Exercise 4.3.15 ([1] Exercise 4.2.16). *Show that the closed surfaces with infinite fundamental group are $K(\pi, 1)$'s.*

Proof. Let M be a closed surface with infinite fundamental group, and let $\tilde{M}$ be its universal cover. Then M is 2-dimensional. Since $\pi_1(M)$ is infinite, $\tilde{M}$ is noncompact. By Poincaré duality, $H_2(\tilde{M}) = 0$. Hence $\tilde{H}_i(\tilde{M}) = 0$ for all i . Hence by proposition 4.1.14, $\tilde{M}$ is contractible. Thus $\pi_i(M) \cong \pi_i(\tilde{M}) = 0$ for all $i \leq 2$. That is, M is a $K(\pi, 1)$. $\square$

Exercise 4.3.16 ([1] Exercise 4.2.19). *If X is a $K(G, 1)$ CW complex, show that $\pi_n(X^n)$ is free abelian for $n \geq 2$.*

Proof. Suppose $\tilde{X}$ is the universal cover of X . By discussion of last exercise, X is contractible, hence $(n-1)$ -connected. Thus X^n is also $(n-1)$ -connected. By Hurewicz theorem 4.1.13, $\pi_n(X^n) \cong H_n(X^n)$ which is free abelian. $\square$

Exercise 4.3.17 ([1] Exercise 4.2.22). *Show that $H_{n+1}(K(G, n); \mathbb{Z}) = 0$ if $n > 1$.*

Proof. We can build a $K(G, n)$ from the Moore space $M(G, n)$ by attaching cells with dimension $> n+1$. Since $H_{n+1}(M(G, n)) = 0$, $H_{n+1}(K(G, n)) = 0$. $\square$

Exercise 4.3.18 ([1] Exercise 4.2.26, [2] 25 autumn T7). *Show that if X is a connected finite-dimensional CW complex with universal cover $\tilde{X}$, then X and $\tilde{X} \times K(\pi_1(X), 1)$ have isomorphic homotopy groups but are not homotopy equivalent if $\pi_1(X)$ contains elements of finite order.*

Proof. Note that

$$\pi_n(\tilde{X} \times K(\pi_1(X), 1)) \cong \pi_n(\tilde{X}) \times \pi_n(K(\pi_1(X), 1)) \cong \pi_n(X)$$

X and $\tilde{X} \times K(\pi_1(X), 1)$ have isomorphic homotopy groups. But since $\pi_1(X)$ contains elements of finite order $\mathbb{Z}_p$, $K(\pi_1(X), 1)$ is infinite dimensional by exercise 2.3.13. And we know that $H_*(K(\mathbb{Z}_p, 1); \mathbb{Z}_p)$ has infinite many nonzero terms. Hence X and $\tilde{X} \times K(\pi_1(X), 1)$ have non-isomorphic homology groups, thus not homotopy equivalent. $\square$

Exercise 4.3.19 ([1] Exercise 4.2.28, [3] 24 SF T5). *Show that the group $\mathbb{Z}_p \times \mathbb{Z}_p$ with p prime cannot act freely on any sphere S^n .*

Proof. Suppose $\mathbb{Z}_p \times \mathbb{Z}_p$ acts freely on S^n , then we have a covering map $S^n \rightarrow M = G/\mathbb{Z}_p \times \mathbb{Z}_p$. Then we have

$$\begin{aligned}\pi_1(M) &= \mathbb{Z}_p \times \mathbb{Z}_p \\ \pi_i(M) &= \pi_i(S^n) = 0, \quad i < n \\ \pi_n(M) &= \pi_n(S^n) = \mathbb{Z}\end{aligned}$$

Hence we can attach cells of dimension greater than n to M to obtain a $K(\mathbb{Z}_p \times \mathbb{Z}_p, 1)$. In particular, we only need one $(n+1)$ -cell. Hence $H^{n+1}(K(\mathbb{Z}_p \times \mathbb{Z}_p, 1); \mathbb{Z}_p)$ is $\mathbb{Z}_p$ or 0. However, note that $K(\mathbb{Z}_p \times \mathbb{Z}_p, 1) = K(\mathbb{Z}_p, 1) \times K(\mathbb{Z}_p, 1)$, and $K(\mathbb{Z}_p, 1)$ is the infinite lens space L_p , and its cohomology is given by

$$H^k(L_p; \mathbb{Z}_p) = \mathbb{Z}_p, \quad \forall k$$

Hence by Künneth formula, we have

$$H^{n+1}(K(\mathbb{Z}_p \times \mathbb{Z}_p, 1); \mathbb{Z}_p) \cong \bigoplus_{i+j=n+1} H^i(L_p; \mathbb{Z}_p) \otimes H^j(L_p; \mathbb{Z}_p) \cong \mathbb{Z}_p^{n+1}$$

which is a contradiction. $\square$

Exercise 4.3.20 ([1] Exercise 4.2.32). *Show that if $S^k \rightarrow S^m \rightarrow S^n$ is a fiber bundle, then $k = n - 1$ and $m = 2n - 1$.*

Proof. By local triviality, $m = k + n$. If $S^k \rightarrow S^m \rightarrow S^n$ is a fiber bundle, we have a long exact sequence

$$\cdots \rightarrow \pi_i(S^m) \rightarrow \pi_i(S^n) \rightarrow \pi_{i-1}(S^k) \rightarrow \pi_{i-1}(S^m)$$

Hence for $i < m$, $\pi_i(S^n) \cong \pi_{i-1}(S^k)$. Thus

$$\begin{aligned}\pi_{i-1}(S^k) &= 0, \quad i < n \\ \pi_{n-1}(S^k) &\cong \mathbb{Z}\end{aligned}$$

Thus for $k = n - 1$. $\square$

Exercise 4.3.21 ([1] Exercise 4.2.34). *Let $p : S^3 \rightarrow S^2$ be the Hopf bundle and let $q : T^3 \rightarrow S^3$ be the quotient map collapsing the complement of a ball in the 3-dimensional torus $T^3 = S^1 \times S^1 \times S^1$ to a point. Show that $pq : T^3 \rightarrow S^2$ induces the trivial map on π_* and $\tilde{H}_*$, but is not homotopic to a constant map.*

Proof. Since the universal cover of T^3 is $\mathbb{R}^3$, hence

$$\pi_i(T^3) = 0, \quad i > 1$$

Note that $\pi_1(S^2) = 0$, hence pq induces the trivial map on π_* . For $\tilde{H}_*$, the only possible nontrivial map is

$$(pq)_* : H_2(T^3) \rightarrow H_2(S^2) \cong \mathbb{Z}$$

But this map factors through $H_2(S^3) = 0$, hence it induces the trivial map on $\tilde{H}_*$.

Suppose there exists a homotopy $F : T^3 \times I \rightarrow S^2$ such that $F(-, 0) = pq$ and $F(-, 1)$ is the constant map. Then since p is a fiber bundle, it has the homotopy lifting property, thus there exists a lifting $\tilde{F} : T^3 \times I \rightarrow S^3$ such that $\tilde{F}(-, 0) = q$ and $\tilde{F}(-, 1)$ is the constant map. This makes a contradiction, since q is a degree 1 map, but the constant map has degree 0. $\square$

Exercise 4.3.22 ([1] Exercise 4.3.1). *Show there is a map $\mathbb{RP}^\infty \rightarrow \mathbb{CP}^\infty$ which induces the trivial map on $\tilde{H}_*(-; \mathbb{Z})$ but a nontrivial map on $\tilde{H}^*(-; \mathbb{Z})$.*

Proof. Since $\mathbb{CP}^\infty$ is a $K(\mathbb{Z}, 2)$, by theorem 4.1.16, we have

$$\langle \mathbb{RP}^\infty, \mathbb{CP}^\infty \rangle \cong H^2(\mathbb{RP}^\infty; \mathbb{Z}) = \mathbb{Z}_2$$

Hence we can choose a map $f : \mathbb{RP}^\infty \rightarrow \mathbb{CP}^\infty$ representing $\mathbb{Z}_2$. That is to say, if we denote a generator of $H^2(\mathbb{CP}^\infty; \mathbb{Z})$ by α , then $f^*(\alpha)$ is a generator of $H^2(\mathbb{RP}^\infty; \mathbb{Z})$, hence f induces a nontrivial map on $\tilde{H}^*(-; \mathbb{Z})$.

On the other hand, we know that $\tilde{H}_k(\mathbb{RP}^\infty; \mathbb{Z})$ is 0 or $\mathbb{Z}_2$ for all k , $\tilde{H}_k(\mathbb{CP}^\infty; \mathbb{Z})$ is 0 or $\mathbb{Z}$ for all k , hence f must induce a trivial map on $\tilde{H}_*(-; \mathbb{Z})$. $\square$

Exercise 4.3.23 ([1] Exercise 4.3.3). *Suppose that a CW complex X contains a subcomplex S^1 such that the inclusion $S^1 \hookrightarrow X$ induces an injection $H_1(S^1; \mathbb{Z}) \rightarrow H_1(X; \mathbb{Z})$ with image a direct summand of $H_1(X; \mathbb{Z})$. Show that S^1 is a retract of X .*

Proof. By universal coefficient theorem,

$$H^1(X; \mathbb{Z}) \cong \text{Hom}(H_1(X; \mathbb{Z}), \mathbb{Z})$$

contains a $\mathbb{Z}$ summand. Since S^1 is a $K(\mathbb{Z}, 1)$, by theorem 4.1.16,

$$\langle X, S^1 \rangle \cong H^1(X; \mathbb{Z})$$

Hence we can choose $f : X \rightarrow S^1$ such that $f^*\alpha = \alpha$, where α is a generator of $H^1(S^1; \mathbb{Z})$. Thus $(f \circ i)^*\alpha = \alpha$. By theorem 4.1.16 again,

$$\langle S^1, S^1 \rangle \cong H^1(S^1; \mathbb{Z}) = \mathbb{Z}$$

Hence $f \circ i$ and id_{S^1} are homotopic. Then there exists a homotopy $F : S^1 \times I \rightarrow S^1$ such that $F(-, 0) = f \circ i, F(-, 1) = \text{id}_{S^1}$. Since $S^1 \rightarrow X$ is a cofibration, F can be extended to $\tilde{F} : X \times I \rightarrow S^1$ such that $\tilde{F}(-, 0) = f$. Then if we take $r = \tilde{F}(-, 1)$, r is a retraction. $\square$

Exercise 4.3.24 ([2] 24 autumn T2). *Solve the following problems:*

(a) *Is it true that every continuous map $f : S^{2024} \rightarrow \mathbb{RP}^{2024}$ is null-homotopic?*

(b) *Is it true that every continuous map $f : S^{2024} \rightarrow \mathbb{CP}^{1012}$ is null-homotopic?*

Proof. It is equivalent to say about $\pi_{2024}(\mathbb{RP}^{2024})$ and $\pi_{2024}(\mathbb{CP}^{1012})$.

(a) Since S^{2024} is the universal cover of $\mathbb{RP}^{2024}$, we have

$$\pi_{2024}(\mathbb{RP}^{2024}) = \pi_{2024}(S^{2024}) = \mathbb{Z}$$

Thus the statement is false, for example, we can take the canonical quotient map $p : S^{2024} \rightarrow \mathbb{RP}^{2024}$, and it is not null-homotopic.

(b) There is a fiber bundle $S^1 \rightarrow S^{2025} \rightarrow \mathbb{CP}^{1012}$, hence we get a long exact sequence

$$\cdots \rightarrow \pi_{2024}(S^1) \rightarrow \pi_{2024}(S^{2025}) \rightarrow \pi_{2024}(\mathbb{CP}^{1012}) \rightarrow \pi_{2023}(S^1) \cdots$$

Thus we get

$$\pi_{2024}(\mathbb{CP}^{1012}) = \pi_{2024}(S^{2025}) = 0$$

So the statement is true.

□

Chapter 5

Miscellaneous

5.1 Basis

Theorem 5.1.1 (Poincaré-Hopf theorem). *Let M be a compact smooth manifold. Let X be a vector field on M with isolated zeroes. Then*

$$\sum_i \text{ind}_{x_i}(X) = \chi(M)$$

In particular, M has a nonvanishing vector field if and only if $\chi(M) = 0$.

Theorem 5.1.2 (Leray-Hirsch theorem). *R is a commutative ring, and $F \xrightarrow{i} E \xrightarrow{p} B$ is a fiber bundle, such that*

- (a) $H^n(F; R)$ is a finitely generated free R -module for all n .
- (b) there exist $c_j \in H^{k_j}(E; R)$ such that $\{i^*c_j\}$ forms a basis for $H^*(F; R)$.

Then there is an isomorphism of R -module:

$$\begin{aligned} \Phi : H^*(B; R) \otimes_R H^*(F; R) &\longrightarrow H^*(E; R) \\ b \otimes i^*c_j &\longmapsto p^*b \smile c_j \end{aligned}$$

5.2 Exercise

Exercise 5.2.1 ([2] 25 autumn T3). *Let X be a compact Lie group of dimension ≥ 1 . Show that the Euler characteristic of X vanishes.*

Proof. Since X admits a nowhere vanishing vector field induced by multiplication, by Poincaré-Hopf theorem 5.1.1, $\chi(X) = 0$. □

Exercise 5.2.2 ([2] 25 autumn T9). *Let X be the space of “a line contained in a plane in $\mathbb{C}^3$ ”, namely a point $F \in X$ is described as a sequence*

$$\{0\} \subset l \subset P \subset \mathbb{C}^3,$$

where P is a complex plane and l is a complex line.

1. *The $\text{GL}(3, \mathbb{C})$ -action on $\mathbb{C}^3$ induces an action on X . Describe the stabilizer of the action and prove that X is a smooth manifold.*
2. *Compute the cohomology ring $H^*(X; \mathbb{R})$.*

Proof. 1. For a point $F \in X$ and $A \in \text{GL}(3, \mathbb{C})$, if A is in the stablizer of F , then $A \cdot l = l$, $A \cdot P = P$. Hence if we choose a basis $\{v, w, u\}$ of $\mathbb{C}^3$ such that

$$l = \mathbb{C}\{v\}, P = \mathbb{C}\{v, w\}$$

then A is represented by an upper-triangle matrix. Conversely, if A is upper-triangle under this basis, then $A \cdot F = F$. Thus the stablizer G is a Lie group consisted of all upper-triangle matrices acting on $\text{GL}(3, \mathbb{C})$ and X is the orbit space $\text{GL}(3, \mathbb{C})/G$. This action is smooth, free, and proper, hence X is smooth manifold with complex dimension

$$\dim X = \dim \text{GL}(3, \mathbb{C}) - \dim G = 9 - 6 = 3$$

2. Consider a map $\pi : X \rightarrow \mathbb{CP}^2$ defined by $\pi(F) = [P]$, then this is a fiber bundle with fiber $\mathbb{CP}^1$. By Leray-Hirsch theorem 5.1.2,

$$H^*(X; \mathbb{R}) \cong H^*(\mathbb{CP}^1; \mathbb{R}) \otimes H^*(\mathbb{CP}^2; \mathbb{R})$$

as $\mathbb{R}$ -modules. Suppose

$$H^*(\mathbb{CP}^1; \mathbb{R}) = \mathbb{R}[\alpha]/(\alpha^2), |\alpha| = 2$$

$$H^*(\mathbb{CP}^2; \mathbb{R}) = \mathbb{R}[\beta]/(\beta^3), |\beta| = 2$$

then

$$H^*(X; \mathbb{R}) \subseteq \mathbb{R}[\alpha, \beta]/(\alpha^2, \beta^3, \alpha\beta - \beta\alpha) := R$$

as a subring. However, R has dimension 1 at grading 0,6; dimension 2 at grading 2,4; dimension 0 at other gradings, which has already coincided with $H^*(X; \mathbb{R})$. Hence $H^*(X; \mathbb{R})$ doesn't have other relations, that is,

$$H^*(X; \mathbb{R}) = R = \mathbb{R}[\alpha, \beta]/(\alpha^2, \beta^3, \alpha\beta - \beta\alpha), |\alpha| = |\beta| = 2$$

□

Exercise 5.2.3 ([3] 25SF T1). *Let $n > 1$ be a positive integer.*

(1) *Is the tangent bundle of $S^{2n} \times S^{2n}$ trivial? Justify your answer.*

(2) *Is the tangent bundle of $S^{2n} \times S^{2n-1}$ trivial? Justify your answer.*

Proof. (1) We compute that

$$\chi(S^{2n} \times S^{2n}) = \chi(S^{2n})\chi(S^{2n}) = 4 \neq 0$$

Hence by Poincaré-Hopf theorem 5.1.1, $S^{2n} \times S^{2n}$ doesn't admit any nowhere vanishing vector field. Thus the tangent bundle of $S^{2n} \times S^{2n}$ is nontrivial.

(2) Note that there exists a nowhere vanishing vector field s on S^{2n-1} . s is a section of TS^{2n-1} , hence TS^{2n-1} can be factorized as $s \oplus E$. Now consider s as the normal bundle of S^{2n} in $\mathbb{R}^{2n+1}$, then $s \oplus TS^{2n}$ is trivial. Hence the tangent bundle of $S^{2n} \times S^{2n-1}$ is trivial.

□

Exercise 5.2.4 ([2] 25SF T4). *Prove that $\mathbb{CP}^2$ does not admit an immersion into $\mathbb{R}^6$.*

Proof. Consider the Pontrjagin class of $\mathbb{CP}^2$:

$$p(\xi) = (1 - a^2)^3 = 1 - 3a^2$$

where ξ is the tangent bundle of $\mathbb{CP}^n$, a is a generator of $H^2(\mathbb{CP}^2)$. If $\mathbb{CP}^2$ can be immersed into $\mathbb{R}^6$, then denote its normal bundle by $\bar{\xi}$, we have

$$p(\bar{\xi}) = p(\xi)^{-1} = 1 + 3a^2$$

But ξ is a bundle of rank 2. Hence $p_1(\xi) = 0$, which is a contradiction.

□

Bibliography

- [1] Hatcher, Algebraic Topology
- [2] Qiuzhen College, Qualifying exams, <https://qzc.tsinghua.edu.cn/syzs/ktzy.htm>
- [3] Yau mathematics contests, <https://www.yau-contest.com/lists-jxxg.html>